

KU EPS 100 NOTES

DEFINITION OF PSYCHOLOGY.

The word psychology comes from the Greek words *psyche* which means —mind/soul/spirit and *logos* meaning —knowledge or study.

Formally defined, psychology is the *scientific study of behavior and mental processes of humans and other animals.*

Psychology as a science.

Psychology is a scientific study.

Psychological research uses *scientific methods* to provide enlightening and reliable information about behavior. Science is derived from the Latin word —scire i.e. to know. Science avoids mere opinions, intuitions and guesses and strives to know by using *objective evidence*. Science uses *logic* to reason about the possible causes of a phenomenon and then *tests* the resulting ideas psychology relies on *empirical evidence* i.e. information gained from direct observation. The information is *collected systematically* through carefully designed experiments; various assumptions can be validated or refuted.

The study of behavior.

Behavior means the *outward (overt)* observable acts of a person/ animal, either alone or in a group.

Study of mental processes

Psychologists also study *covert* behaviors. These are *private, internal activities*, such as *thinking; dreaming; remembering; perceiving; reasoning; problem solving* and other mental events.

Study of humans and other animals

THE GOALS OF PSYCHOLOGY

- i) **Describe behavior:** Answering psychological questions requires a careful description of behavior. This is based on making a detailed record of behavioral observation.
- ii) **Understanding behavior:** Explaining an event-Stating the causes of behavior and determining how they led to the described behavior.

□ E.g. research on —bystander apathy‖ reveals that people often fail to help when other possible helpers are nearby. Why? Because a —diffusion of responsibility‖ occurs. Basically, no one feels personally obligated to assist. As a result, the more potential helpers there are, the less likely it is that anyone will help (Darley 2000; Darley and Latane, 1968). This can help us explain a perplexing problem.

- iii) **Predicting behavior:** This is the ability to forecast/ foretell behavior accurately. e.g.

- How people learn most effectively.
- Bystander apathy – makes a prediction of the chances of getting help- (having many potential helpers is no guarantee that anyone will help).

- iv) **Controlling or influencing phenomena:** This refers to altering conditions that affect behavior e.g. if you suggest changes in a classroom that help students learn better, you have exerted control. If you help a student overcome terrible exam/ test anxiety, control is involved.

N.B. Psychological control must be used wisely and humanely.

MAJOR PERSPECTIVES FOLLOWED IN THE STUDY OF PSYCHOLOGY

A perspective is a broad look at a topic/ issue.

A number of perspectives have been used to explain human thought and behaviour.

i) **The Psychoanalytic perspective.**

The psychodynamic perspective originated with the work of Sigmund Freud. This view of psychology and human behavior emphasizes the role of the unconscious mind, early childhood

experiences, and interpersonal relationships to explain human behavior and to treat people suffering from mental illnesses. He viewed psychological traits and disorders as having an origin in sexual and aggressive drives ii)**The Behavioural perspective(early 20th century)**

- Behaviour theorists include B.F. Skinner and J.B. Watson.
- Proposes that behaviour is a product of learning
- It strove to make psychology a more scientific discipline by focusing purely on observable behavior.
- Today, the behavioral perspective is still concerned with how behaviors are learned and reinforced.
- Behavioral principles are often applied in mental health settings, where therapists and counselors use these techniques to explain and treat a variety of illnesses

iii)The biological perspective

- This helps us understand how the body and the brain work in unison to create emotions, memories and sensory experiences.
- This perspective studies evolutionary and hereditary influences on behaviour, how messages are transmitted within the body, and how blood chemistry is related to moods and motives.

iv)The humanistic perspective

- This school of thought known as humanistic psychology emerged during the 1950s.
- This perspective emphasizes the role of motivation on thought and behavior.
- It emphasizes that the human being has the capacity to deliberately choose a meaningful life pattern and therefore grow to greater maturity and fulfillment.
- It was influenced greatly by the work of prominent humanists such as Carl Rogers and Abraham Maslow.
- Concepts such as self-actualization are an essential part of this perspective.

v)Cognitive Perspective

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- This perspective began during the 1960s.
- ‘_Cognition’ is the ability to utilize the mind in terms of mental processes such as memory, thinking, problem solving, language and decision-making.

- In this perspective, psychologists explore the way human beings use their mental processes to reason and solve problems.
- Influenced by psychologists such as Jean Piaget and Albert Bandura, this perspective has grown tremendously in recent decades.

iv) Cross-cultural perspective

- This is a fairly new perspective in which psychologists and researchers look at human behavior across different cultures.
- It looks at cultural and ethnic similarities and differences on both psychological and social functioning.
- The assumption is that each person has a cultural and ethnic background which influences their thinking and behavior depending on the environment in which they are.

GROWTH OF PSYCHOLOGY

NB: The roots of psychology lie in philosophy (use of logic and specialization to understand the nature of reality, experience, and values) on the one hand and physiology on the other (Study of biological workings of the body, including the brain).

Philosophers such as Plato, Aristotle, Descartes and Locke – raised provocative questions about human thoughts, feelings and behavior. E.g.

- Renes Descartes (French) focused attention on the distinction between mind and body. ▪

John Locke (English philosopher) stressed that all human knowledge arises from experience of the entire world and from reflection about it.

Among the early scientific psychologists were the structuralists.

Approaches to Psychology/schools of thought

A school of thought comprises a system of ideas that a proponent wishes to promote. Theories are based on schools of thought.

1. Structuralism

Wilhelm Wundt (1832 – 1920): Founder of scientific psychology.

- Set up the first psychology laboratory in 1879 in Leipzig, Germany.
- He defined the task of psychology as the systematic study of the structure of the conscious adult mind.
- Structuralism sought to ***break experiences into basic elements or structures***, in the same way that a substance like water could be broken into molecules of hydrogen and oxygen.
- He sought to identify the ***building blocks of consciousness*** i.e.
 - i) ***Sensations*** – which arise from the eyes, ears and other sense organs.
 - ii) ***Feelings*** e.g. fear, anger and love
- The goal of structuralism was to describe the rules that determine how particular sensations or feelings may occur at the same time or in sequence, combining in various ways into mental structures.
- Wundt was primarily interested in ***selective attention*** – the process by which we determine what we are going to attend to at any given moment.
- He borrowed a tool of philosophy i.e. ***introspection*** (looking inward) for studying mental processes.
- One important product of the Leipzig laboratory was its students, who carried new, scientific psychology to the universities in other countries. i.e.
 - Stanley Hall – established the first American Psychology Laboratory at John Hopkins University in 1883.
 - J.M. Cattell – first American to be called a professor of psychology at the University of Pennsylvania in 1888.
 - British born Edward Bradford Titchener who went to Cornell University (USA). He broke down consciousness into 3 basic elements:
 - physical ○ feelings ○ images

Limitations

- If an experience is broken into basic elements, it may not retain its essential character when subjected to this reductionist approach.

- Introspection – often altered the nature of the conscious mental processes (analyzing what you are experience changes the experience.)
- The observation of structuralists could not be objectively repeated with the same results (results were different.)

Contributions

- The study of perception and sensation continues to be a part of contemporary psychology.
- Looking inward (introspection) is still used in studies of hypnosis, meditation, problem – solving, moods e.t.c

2. Functionalism

- Adherents of functionalism sought to understand how our *minds help us to adapt to the world around us* i.e. *how to function* in it.
- Functionalists were strongly influenced by **Charles Darwin (1809- 1882)**, whose theory of evolution by natural selection stressed that some organisms possess *characteristics that enable them to survive* and reproduce more fruitfully than others (Survival for the fittest). Darwin called these *characteristics adaptations*.
- Functionalists applied Darwin's theory to mental characteristics.
- E.g. **William James (1842 – 1910)**, who set up the first psychology laboratory in the USA at Harvard University, studied the ways in which consciousness helps an individual survive and adapt to the environment.
- He wrote a book – —*Principles of Psychology* in 1890, which details his view of the nature of psychology.
- Functionalists hold the view that psychological states or processes, such as consciousness evolved because they serve particular functions, such as guiding the activities of the individual.
- They wanted to find out how various *mental processes, such as perception, learning, thinking, help people to adapt and survive*.
- To accomplish this purpose, they continued to use introspection in their research. However, they also introduced another research method: collecting data from **observation** of human and animal behavior.

Contributions

- Their emphasis on Darwin's theory of natural selection and its links between human and non – human animals led them to theorize that human psychology is related to the psychology of animals. Observation of animals could provide clues to human behavior.
- Focus on issues of society e.g. improving methods of education. □ Expanded the data of psychology to include observations of behavior.

iii) Evolutionary approach

- Based on Charles Darwin's book —The Origin of Species
- Darwin argued that the forms of life we see today are the result of **evolution**- of changes in life forms that occur over many generations.
- He said that evolution occurs through natural selection, which promotes survival of the fittest individuals.
- Most evolutionists today see natural selection operating at the level of genes.
- Genes that result in characteristics and behaviours that are adaptive and useful in a certain environment will enable the -creatures that inherited them to survive and reproduce, thereby passing those genes on to the next generation.
- Genes that result in characteristics that are not adaptive in that environment are not passed on to subsequent generations, because the creatures possessing them do not survive to produce.
- Central to the evolutionary psychology approach is the idea that ***certain cognitive strategies and goals are so important that natural selection has built them into our brains.***
- It asserts that the patterns of behaviour seen in a species are products of evolution.
- It therefore focuses on the ***evolutionary origins of behavior patterns and mental processes, exploring what adaptive value they have or had and what function they serve or served in our emergence as a distinct species.***
- They ask – How do human beings get to be the way they are?
- They study such diverse topics as perception, language, helping others (altruism), parenting, happiness, sexual attraction and mate selection, jealousy and violence.
- By studying such phenomena in different species, different habitats, different cultures, and in males and females, evolutionary psychologists ***seek to understand the basic programs that guide thinking and behavior.***

- They believe that general cognitive strategies are inborn.
- They see the *mind as “hardwired”*, so that human beings are *predisposed to think and act in certain ways*.
- They contend that these fixed programs evolved hundreds of thousands of years ago among our ancestors.
- Hence many of the genes we possess today are the result of natural selection.
- The human mind was sculpted by natural selection, and it is this evolved organ that constitutes the subject matter of psychology.
- Their approach assumes that the behaviour of animals and humans today is due to evolution through natural selection. E.g.
- Aggression- form of territory protection
- Gender differences- male selection preferences
- Helping one's genes survive in future generations

iv) Psychodynamic approach: Sigmund Freud 1856- 1939

- Rooted in *Freud's psychoanalysis*
- He was a Viennese physician, specializing in neurology (study of treatment of diseases of the brain and nervous system).
- He proposed that the mind is made up of the id, ego, and the superego
- He believed that *mental life is like an iceberg* – a small part is expressed to view.
- He called the area of the mind that lies outside of personal awareness the *unconscious*.
- He assumes that *behaviour and mental processes reflect* constant and mostly *unconscious psychological struggles* raging within each person.
- He viewed the unconscious as a *dynamic cauldron of primitive sexual and aggressive drives, forbidden desires, nameless fears and wishes and traumatic childhood memories*. Anxiety, depression, or other disorders are seen as overt signs of inner turmoil. Although repressed (or hidden from awareness), unconscious impulses press on the conscious mind and find expression in disguised or altered form, including dreams, mannerisms, slips of the tongue and symptoms of mental illness as well as in socially acceptable pursuits such as art and literature
- To uncover the unconscious, Freud developed the **technique of free association**, in which the client lies on a couch, recounts dreams and says whatever comes to mind.

Contributions:

- Freud came up with a developmental approach to an individual's personality (development proceeds through stages). It recognized the significance of early childhood experiences as affecting later behavior. (e.g. feeding, toilet training). First years of life help shape adult personality.
- He recognized the significance of motivation
- He recognized the importance of anxiety as an antecedent for behavioral disorders
- Laid a foundation for the study of personality and psychological disorders
- His notion of the unconscious, with its irrational urges and drives provided insight into understanding the emotional lives of people. It encouraged psychologists to consider the impact on behavior of processes not immediately available to conscious inspection
- He helped to legitimize the study of human sexuality
- Took a strong stand on determinism i.e. nothing is an accident. If we probe deeply enough, we will find the causes of every thought and action
- Interpretations of the various forms of art and literature fall within a Freudian framework
 - Created psychoanalysis – the first —talking cure

Criticisms:

- Unverified theoretical constructs – some of his theories and hypothesis could not be put to test e.g. structure of personality psychic energy
- Unreliable data e.g. from free association
- Fictional concepts e.g. Oedipus complex, castration anxiety
- He made the disconcerting suggestion that people are not masters of their own mind (behaviour is governed by unconscious forces)
- Controversial proposal that behaviour is greatly influenced by how people cope with their sexual urges

v) Behaviorism

- Founded by **John B. Watson (1878-1958)**.
- For Watson, psychology was the study of *observable measurable behavior*
- He stressed the **stimulus – response (S-R)** approach to psychology. He simply observed the relationship between stimuli (events in the environment) and an animal's responses
- Watson adopted Russian physiologist **Ivan Pavlov's (1849-1936)** concept of **conditioning** to explain most behavior.

- Ivan Pavlov noted that the dogs in his laboratory began to salivate as soon as they had their feeder coming, even before they could see the food. He decided to find out whether salivation, an automatic reflex, could be shaped by learning. He began by repeatedly pairing the sound of a buzzer/bell with the presence of food. Next he observed what happened when the buzzer was sounded without introducing food. After repeated pairings, the dogs salivated to the buzzer alone. He called this simple form of training **conditioning**. The new school of psychology was inspired by **observation, followed by rigorous experiment**.
- A conditioned response is a learned reaction to a particular stimulus. Watson eagerly claimed: *“ Give me a dozen healthy infants, well-formed and my own special world to bring them up in and I’ll guarantee to take any one at random and train him to become any type of specialist I might select – doctor, lawyer, artist, merchantchief, and yes beggar man and thief” (Watson, 1913/1994).*
- Watson attempted to demonstrate that all *psychological phenomena* - even Freud’s unconscious motivations – are the **result of conditioning**.
- In one of his experiments, he attempted to **create a conditioned fear response** in a two month old boy-**Little Albert**. Albert visited Watson’s laboratory and was delighted by a tame, furry white rat, but became visibly frightened when Watson banged a steel bar with a hammer just behind the infant’s head. On the second visit, Watson placed the rat near Albert and the moment the baby reached out and touched the rat, Watson banged the hammer. After half a dozen pairings, little Albert began crying the instant the rat was introduced, without any banging. Further experiments found that Albert was frightened by anything white and furry- rabbit, dog, coat, cotton wool e.t.c. i.e. had generalized .Watson also demonstrated that fears could be eliminated by conditioning (reconditioning).
- One of the best known behaviorists **B.F. Skinner (1904-1990)**, fervently believed that psychologists should only study **observable and, measurable** behavior. He added a new element to the behaviorist’s repertoire of **reinforcement**. If the consequences of behavior are reinforcing, we are likely to repeat the behavior. Conversely, if a behavior produces an undesirable outcome (punishment), we are less likely to do it again.

Contributions:

- Skinner's work in operant conditioning shows the effects of reinforcement on behavior
- Skinner has extensive writings on language learning, programmed instruction, philosophy of science and politics
- Emphasis on objective observation
- Behaviorists insights have improved psychotherapy and education

Criticisms:

- Many behaviorists objected to the study of mental processes. These objections have been refuted by subsequent research.

vi) Gestalt psychology

- Gestalt psychology – is taken from the German word Gestalt, which means —**whole**” or —Form
- It emphasized the *overall pattern of thoughts or experience*.
- It was pioneered by a group of psychologists in Germany i.e.
 - *Max Wertheimer (1880-1943)*
 - *Wolfgang Kohler (1887-1967)*
 - *Kurt Koffka (1886-1941)*
- Gestalt psychologists studied thinking, learning and perception as whole units, not by analyzing experiences into parts.
- Their slogan was —**The whole is greater than the sum of its parts** e.g.
 - If we look at a tree we just that, a tree, not a series of isolated leaves and branches
 - A flight of birds – focus is on the whole flock, not individual birds. The flock is the perceptual unit.
- It is a mistake according to Max Wertheimer, to analyze psychological events into pieces or elements as the structuralists did. Like a melody (made up of musical notes put together), many experiences cannot be broken into smaller units.

Contributions

- Studies of perception have been influenced by the Gestalt viewpoint. Perception is our gateway to the world; if our perceptions are not accurate, our corresponding thoughts and feelings will be based on a distorted view of reality.
- The research of the Gestaltists addressed how the brain works, and today Gestaltism has become integrated into studies of the brain itself.
- Has also paved way for the modern study of personality.

vii) Humanistic psychology/Phenomenological approach ➤

Proponents □ **Abraham Maslow (1908 – 1970)** □ **Carl Rogers (1902 – 1987)**

- According to humanistic psychology:
 - People have positive values
 - Free will
 - Deep inner creativity which allows them to choose life – fulfilling paths to personal growth.
- It emphasizes: Love, creativity, belonging, self – esteem, self – expression, spirituality, and self – actualization (developing one’s potential fully and becoming the best person possible.)
- Humanists also stress more subjective factors such as:
 - Self – image (perception of your body, personality and capabilities.)
 - Self- evaluation (appraising yourself as good or bad.)
 - Frame of reference (mental perspective used to interpret events.)
- Humanists seek to understand how we perceive ourselves and experience the world.
- Behaviour is determined by *each person’s capacity to choose how to think and act.*
- These choices are driven by each *individual’s unique perception of the world.* E.g. if you see the world as a friendly place, you are likely to be optimistic and secure. If you perceive it as hostile, you are likely to be defensive and fearful.
- Proponents of this approach argue that behaviour and mental processes can be fully understood only by understanding the perceptions and feelings of individuals.

- Humanistic psychologists also believe that people are essentially good, that they are in control of themselves, and that they have an innate tendency to grow toward their highest potential.

Contributions

- Optimistic - views the outlook for personal fulfillment optimistically and positively
- It puts forward the notion that man is basically good and has tremendous possibilities for self – fulfillment.
- Has increased psychologists' awareness of the importance of such things as love, feeling needed, personal fulfillment and self – esteem.

Criticisms

- Reliance on non – scientific approach to understanding human behavior. The theories lack any empirical validity.
- Uses introspection, which is unreliable. **viii)The cognitive approach**

➤ Emerged in the 1950's and 1960's. ➤ Proponents:

- **Herbert A. Simon** (Computer psychologist/ scientists) □ **Allan Newell.**
- **Noam Chomsky – linguist**
- **Gardener (1985)**

- It emphasizes the study of our **mental processes** in the broadest sense: thinking, feeling, learning, remembering, and making decisions and judgments e.t.c.
- Focuses on how we take in, mentally represent, and store information; how we perceive and process that information, and how cognitive processes are related to our behavior.
- Focus on the above to understand individual and social behaviours, from decision making and problem solving to interpersonal attraction and intelligence.
- Study also the building blocks of cognition and determine how they produce complex behaviours such as remembering, writing a word or making a decision.
- Cognitive psychologists have begun to address questions about the neurological mechanisms that underlie such cognitive processes as learning, memory, intelligence and emotion giving rise to the rapidly expanding field of cognitive neuroscience. It blends cognitive psychology and neuroscience (the study of the brain). Cognitive neuroscientists argue that —**the mind is what**

the brain does and therefore hope to discover the nature, organization and operation of mental operations by studying the brain.

ix) Positive psychology

- Another emerging perspective is positive psychology.
- It focuses on **positive experiences**, including: subjective feelings of happiness and well being, development of such individual traits such as intimacy, integrity, leadership, altruism and wisdom, kinds of families, work settings and communities that encourage individuals to flourish, nurturance of mental wellness, and relationship between positive emotions and physical health. Emphasis is on **building of positive qualities**

x) Individual Differences

- The individual differences school of thought became prominent after 1930s.
- Its main proponent was Sir. Francis Galton.
- Galton asserted that behaviour could be inherited.
- He justified this by giving results of a study on his family tree. He was generally impressed by the great number of exceptional people in his family, for example, geniuses like Charles Darwin (his half cousin).
- Galton is accredited with formulation of mental tests and the study of individual differences. He is the first person to introduce statistical concepts in the field of psychology, one of them being the concept of **correlation coefficient**. A correlation coefficient is a numerical index used to show the degree of similarity between two or more sets of data.

CONTEMPORARY PSYCHOLOGY

- It is not dominated by any single theoretical perspective.
- Contemporary psychologists tend to see the different perspectives as complementary e.g. when they study aggression, psychologists no longer limit their explanations to the behavioral view (aggressive behavior is learned); or the Freudian perspective (aggression is an expression of unconscious hostility) but also use other theories e.g. evolutionary and cognitive psychology (**Eclectic approach**)

BRANCHES/ FIELDS/SPECIALTIES OF PSYCHOLOGY

- i) **Cognitive psychology:** Focuses on processes such as thinking, attention, perception, decision making, reasoning, concept formation, memory, language,

problem- solving, creativity and related aspects of mental processes. It is interested in one's *internal representation or cognitive maps*. **ii) Developmental psychology:** Focuses on human growth and development throughout the life cycle i.e. from conception to old age.

- Child psychologists – focus on infants and children. They are concerned with issues such as whether babies are born with distinct personalities and temperaments, how infants become attached to their parents and caregivers e.t.c.
- Adolescent psychologists – specialize in the teenagers. Concerned with puberty, changes in relationships with peers and parents and the search for identity e.t.c.
- Life-span psychologists – focus on adult life years. Are concerned with different ways that individuals adjust to partnership and parenting, middle-age, retirement and eventually death.

iii) Social psychology: Concerned with understanding the impact of social environments on the individual.

Are interested in: attitude formation and change; social perception; conformity; social roles; prejudice and discrimination; interpersonal perception (forming impressions of others); interpersonal attraction; and aggression. **iv) Personality psychology:** Explore the uniqueness of the individual; describe the key elements that provide the foundation for human personalities; and investigate how personality develops, and evolves.

v) Experimental psychology: Primarily conduct research on basic psychological processes – in an attempt to understand the nature and causes of behavior e.g. emotion, sensation, motivation, cognition, perception, learning, memory e.t.c. **vi)**

Physiological psychology/ neuroscience: Investigates the biological basis of human behavior, thoughts, emotions, memory, disordered behaviors, e.t.c. Biological psychologists explore such things as the association between behavior and drugs, hormones, genes and brain processes.

vii) Clinical and counseling psychology: Involved in the diagnosis and treatment of psychological problems including such things such as developmental disorders, substance abuse, relationship difficulties, vocational and educational problems and behavior disorders.

□ **Counseling psychologists:** Focus on less serious problems of adjustment e.g. Social or academic adjustment; guidance in career decisions.

□ **Clinical psychologists:** Likely to work in *mental health clinics, mental hospitals, juvenile and adult courts, medical schools and prisons*. They have PhD training in psychological theory, research methods, and techniques of clinical diagnosis and psychotherapy strategies. In their perspectives on causes of psychological problems, they tend to emphasize psychosocial causes e.g. inappropriate learning, faulty attitudes and disturbed interpersonal relationships. They focus on psychotherapy as the best road to improvement.

viii) Educational and school psychology:

Educational psychology – involves the study of learning and teaching methods, focusing on ways to improve educational curricula and training teachers. *School psychology* – concerned with educating and resolving learning and emotional problems. Often administer and interpret personality, interest and ability tests. Are a valuable resource both for troubled students and for concerned teachers trying to cope with the stresses of classroom problems.

ix) Industrial/ organizational psychology: Uses psychological concepts to make the workplace a more satisfying environment for both employees and management. They work with businesses either as company employees or as consultants, designing programs to: improve group morale; increase job satisfaction; foster better communication within the corporation; enhance productivity; increase workers' involvement in decision making; and improving working conditions. They are also concerned with issues such as designing job- training programs and selection of personnel, most suitable for a particular job.

- x) **Engineering psychology/ human factors psychology:** Focuses on the creation of optimal relationships among people, the machines they operate and the environments in which they work. e.g. designing the lighting and instrumentation within the cockpits of sophisticated aircraft to maximize pilot efficiency. In space programs – they have helped to develop optimal/ functional efficiency within the severely limited confines of spacecraft.
- xi) **Health psychology:** Concerned with the relationship between behavioral factors and physical health. It involves the use of psychological principles to promote change in peoples' attitudes and behaviour about health and illness.

Deal with diverse areas e.g.

- Assessing the psychological and physical effects of stress and stress reduction.
- Coping strategies for dealing with serious or catastrophic illness.
- Evaluating the impact of psychological factors on diseases such as cancer and cardiovascular illness.
- Devising ways to test people for susceptibility to disease.
- Seeking to identify the factors that motivate people to engage in life- threatening activities e.g. smoking, overeating, under eating.
- Using psychological theories and interventions to prevent damaging behaviours and to change to health related behaviours(healthy diet, health checks, self-examination, teethbrushing, washing of hands etc)
- Health-related cognitions-investigating the processes that can explain, predict, and change health and illness behaviours
- Processes influencing health-care delivery eg communication, facilitating adherence eg taking medication, preparing for stressful medical procedures

- Psychological aspects of illness: looking at the psychological impact of acute and chronic illness on individuals, careers and families.

xii) Positive psychology: Study of human behavior aimed at discovering and promoting the positive strengths and attributes that enable individuals to thrive and succeed, to be self – fulfilled and happy. (positive qualities of an individual: Optimism; courage; work ethics; interpersonal skills; social responsibility; capacity for pleasure; insight e.t.c.

xiii) Forensic psychology: Application of psychological principles and methods to the criminal justice system. It works with the legal, court and correctional systems:

- to develop personality profiles of criminals/offender profiling;
- to select the jury;
- present evidence;
- give eyewitness testimony;
- make decisions about disposition of convicted offenders e.g. 1991 – Jeffrey Dahmer murdered, dismantled and apparently ate selected body parts from numerous victims;
- help law enforcers understand behavioral problems e.g. family conflict, substance abuse;
- devise treatment programmes e.g. anger management;
- crime prevention; and
- assess the risk of releasing prisoners.

xiv) Artificial intelligence and connectionism

Artificial intelligence: Researchers develop computer models to simulate human cognitive processes e.g. perceive stimuli, problem – solving, learning and making decisions. Artificial intelligence has been successfully applied to such varied pursuits as the diagnosis of illness and location of deposits of valuable resources such as oil.

Connectionism: Focuses on discovering the neurobiological mechanisms underlying learned associations. It helps to study complex human abilities e.g. learning, problem – solving and perception.

- xv) **Sports psychology:** Works with athletes to help improve their performance by helping them learn to concentrate better, deal with stress and practice more efficiently.
- xvi) **Individual differences:** This is concerned with the ways in which people differ from one another including:
 - ❖ Personality
 - ❖ Intelligence
 - ❖ Psychological abnormality
- xvii) **Cross-cultural psychology:** examines the role of culture in understanding behaviour, thought, and emotion. Cross cultural psychologists compare the nature of psychological processes in different cultures, with a special interest in whether or not psychological phenomena are universal or culture specific.

PSYCHOLOGY AS A SCIENCE

- Psychologists gather information by thinking critically about, and scientifically studying the mind/mental processes and behaviour.
- This entails the use of scientific research (objective, systematic, and testable).
- Scientific research is based on the scientific method, an approach that can be used to discover accurate information.

NB: Psychology differs from common sense in various ways:

- + Psychology is a scientific inquiry based on thorough research. Common sense is information gathered from everyday informal observations, knowledge and experiences.
- + Psychology involves judgments which have been rigorously tested and verified in order to ascertain the truth unlike common sense.
- + It relies on objective evidence-It is based on observations, theories, and experiments.
- + It uses logic to reason about phenomena and derive conclusions
- + It relies on empirical evidence
- + Information is gathered systematically
- + It relies on precision, accuracy, and lack of bias
- + Assumptions are validated or refuted

- ✚ Common sense does not use analytical methods, which are crucial to decision making and judgment
- ✚ Psychology rules out the metaphysical but common sense allows us to believe that ghosts and angels are causal agents of events in our lives.
- ✚ Clarity and precision-commonsense notions about behaviour tend to be vague and ambiguous. In contrast, the scientific approach requires that people specify exactly what they are talking about.
- ✚ Scientists subject their ideas to empirical tests. They also scrutinize one another's findings with a critical eye. They demand objective data and thorough documentation before they accept ideas. In contrast, commonsense analyses involves little effort to verify ideas or detect errors.
- ✚ The scientific approach yields more accurate and dependable information than casual analyses and armchair speculation.

Steps in a scientific investigation

a) Conceptualizing a problem/specifying a problem

- This is the question one wants to answer, or the puzzle one wants to solve. It entails narrowing down and zeroing in on what one wants to study.
- Theories can be used in conceptualizing a problem (a theory is a coherent set of interrelated ideas that helps to make predictions and explain data).

b) Formulating a testable hypothesis

- A hypothesis is a tentative idea that might explain a set of observations.
- May also be defined as specific assumptions or predictions that can be tested to determine their accuracy.
- A hypothesis is also a statement of the possible differences, relationships or causes between two or more variables or concepts.
- Hypotheses are derived from existing theories, previous research, or personal observations and experiences.

- To be testable, hypotheses must be formulated precisely, and the variables under study must be clearly defined.

NB: Types of hypothesis

i) Alternative hypothesis

✚ States that there are significant differences or relationships between variables.

✚ It is denoted by H_1

✚ Alternative hypothesis are of two types:

- ✓ ***Directional alternative hypothesis***-shows the effect of the independent variable on the dependent variable eg there is a positive and significant relationship between the qualification of teachers and students' achievement in science subjects
- ✓ ***Non-directional alternative hypothesis***-does not give the direction of effect e.g. there are differences between boys and girls; attitude affects performance etc

ii) Null

hypothesis

✚ It is denoted by H_0

✚ It is stated to show that there is no effect/difference/relationship between the variables

✚ Any difference or relationship is merely due to chance or error.

NB: Researchers achieve these clear formulations of hypotheses by providing operational definitions of the relevant variables.

c) Operational definition of terms

- It establishes precisely what is meant by each variable in the context of the study.
- It makes concepts clearer by being very specific about what observational measurements define them.

d) Select the research method and design of the study

- The research method chosen will depend to a large degree on the nature of the question under study.

- The various methods-experiments, case studies, surveys, naturalistic observation etc have advantages and disadvantages.
- The researcher has to ponder the pros and cons and then select the strategy that appears to be the most appropriate and practical.
- Once researchers have chosen a general method, they must make detailed plans for executing their study-time, number of people needed, where the participants will come from etc (participants or subjects are the persons or animals whose behaviour is systematically observed in a study).

e) Collect the data

- Researchers use a variety of data collection techniques, which include observation, questionnaires, interviews, psychological tests, physiological recordings (blood pressure, muscle tension, brain activity) examination of archival record(census, economic, medical, legal, educational, and business records) etc

f)Analyze the data

- Researchers analyze the raw data to decide whether their hypotheses have been supported or not.
- Statistics play an essential role in the scientific enterprise-in understanding what the data mean.

g)Draw conclusions

- Based on their statistical analysis, conclusions are drawn.

h)Report the findings

- Researchers should share their findings with one another and with the general public.
- They should therefore write up a concise summary of the study and its findings.
- Typically they prepare a report that is delivered at a scientific meeting and submitted to a journal for publication.

RESEARCH METHODS

1) Experimental research

- ❖ It allows psychologists to determine behavior's causes.
- ❖ It's a research method in which an investigator manipulates a variable under carefully controlled conditions and observes whether any changes occur in a second variable as a result.
- ❖ It is a research method in which a group of participants receive a treatment and another group does not. Participants in the experimental group receive the treatment whereas participants in the control group do not. The participants are then observed to determine whether the treatment makes a difference in their behavior.
- ❖ There is an *independent variable (IV)* whose presence is manipulated by the experimenter so that its effects can be determined.
- ❖ The *measured result* is called a *dependent variable (DV)*. Its presence presumably depends on the independent variable.
- ❖ Subjects should be assigned to the experimental or control group on a chance or *random basis* in such a way that each person has an equal chance of being placed in any group (random assignment).
- ❖ This technique is appropriate in establishing cause-effect relationships
- ❖ For example to determine whether T.V. violence causes aggressive behavior- T.V. violence would be considered as the IV and the child's behavior will be the DV.

Advantage

- This approach comes as close as none other known in establishing causality-permits conclusions about cause and effect relationships between variables.

Disadvantages

- A number of variables are not amenable to experimental manipulation e.g. age, gender, etc.

- It would be unethical to manipulate some variables e.g. withholding treatment from a patient and exposing him/her to danger
- Experimental designs are difficult to conduct in natural settings/they are often conducted in an artificial environment. Subjects may react to artificiality and not perform normally.

Cont. Research methods

2) Observation

- ❖ Whereby ongoing behaviour of individuals is recorded with as little interaction between the observer and subjects as possible.
- ❖ Forms:
 - i) **Naturalistic** observation where researchers observe people in real life settings. They do not try to alter behaviour or the environment; they simply record what they see.
 - ii) **Laboratory** observation: researchers observe and record behaviour in a controlled environment, such as a laboratory.
- ❖ Both kinds of observation provide valuable descriptions of behaviour.

Advantages

- Facilitates observation of subjects in their natural settings.
- Development can be observed in various stages.
- Crucial for determining how children/adults function in their everyday life.
- Researcher does not impose himself/herself on the subject.

Disadvantages

- Observation does not explain why people behave the way they do, though the observers may suggest interpretations (does not explain cause-effect/why events are happening).
- An observer's presence can alter behaviour. When people know they are being watched, they may act differently.

- There is a risk of observer bias: the researcher's tendency to interpret data to fit expectations or to emphasize some aspects and minimize others.
- There are ethical problems if consent is not obtained.
- Data obtained from observation is vulnerable to bias and distortion.
- Emotions, prejudices and values may influence the way that behaviours and events are observed.
- Observation may also be time consuming.
- Does not allow for experimental control.
- Does not provide information about the internal psychological events and experiences of subjects, their thoughts, beliefs, fantasies etc
- May be inefficient in studying behaviour if it occurs infrequently or irregularly in the subject's life.
- One may have to wait for a long time before the behaviour of interest is emitted.
- The information got may not be generalized to other subjects.

3)Case study

- ❖ It is an in depth study of an individual, a group of individuals or an institution
- ❖ A case study is frequently used when there is a new phenomenon about which not much is known or very rare events in which few subjects can be found.
- ❖ Case studies provide significant amounts of descriptive information and they can also provide some explanatory information about why as well as what.
- ❖ One can use interviews, direct observation of the subject, examination of records, and psychological testing to collect the data.

Advantages

- They offer useful in- depth information
- Useful in exploring behavior
- Flexibility i.e. the researcher is free to explore avenues of inquiry that arise during the course of the study

Disadvantages

- Time consuming
- Can be quite costly
- Subject drop-out may occur if the study is carried out over an extended period
- They may provide information about the development of a single person, but not how the information applies to general people
- Case studies cannot explain behavior with certainty or make strong causal statements because there is no way to test their conclusions
- Case studies are highly subjective-it is easy for investigators to see what they expect to see in case study research (researchers may focus selectively on information that fits with their expectations).

4)Correlational studies

- ❖ A correlational study seeks to determine whether a relationship exists between variables/ phenomena.
- ❖ Correlations are expressed in terms of direction i.e. positive or negative and magnitude (degree)
- ❖ Two variables that are positively correlated increase or decrease together. For example, studies show a positive or direct correlation between televised violence and aggression. That is, children who watch more violent television tend to fight more than children who watch less violent television.

- ❖ Two variables have a negative or inverse correlation if as one increases the other decreases. Studies show a negative correlation between amount of schooling and the risk of developing dementia (mental deterioration) due to Alzheimer's disease in old age. In other words, the less education, the more dementia.

Advantage

- Correlations enable us to predict one variable in relation to another

Disadvantage

- Correlational research does not allow us to place clear —cause‖ and —effect‖ labels on variables. To investigate cause and effect, researchers turn to the experimental method.

5) Surveys

- Researchers use questionnaires or interviews to gather information about specific aspects of participants' behaviour.
- The survey questions should be unambiguous and clear.
- The people surveyed must be selected with great care.

Advantages

- Surveys are a relatively inexpensive way to collect a lot of data fairly quickly.
- They provide data that can be used to formulate or test a hypothesis.
- They make it relatively easy to collect data on attitudes, opinions, beliefs, preferences etc from large samples of subjects
- Surveys are often used to obtain information on aspects of behaviour that are difficult to observe

Disadvantages

- They depend on self-report data-intentional deception can distort participants' verbal reports about their behaviour.
- The value of survey is limited by what people are capable of reporting accurately.

6) Clinical method

- ❖ It involves a researcher and a subject at a time
- ❖ It entails observation and careful questioning and allows for flexibility

Advantage

- Questions can be individualized for each subject
- It facilitates a detailed observation/questioning

Disadvantages

- Conclusions arrived at depend on the interviewer's questioning ability
- Results obtained may require verification by other researchers

HEREDITY AND ENVIRONMENT

The determinants of human development

- Although genes play a role in human behaviour, they alone do not determine who we are.
- Genes exist within the context of a complex environment and are necessary for an organism to exist.
- The interaction of nature and nurture, genes and environment, influences every aspect of mind and behaviour. Neither factor operates alone.
- Heredity and environment operate together to produce a person's intelligence, temperament, height, weight, personality, abilities etc

Environment

- Refers to all the surrounding conditions and influences that affect the development of living things.
- It includes the food we eat, the air we breathe, and the many different physical and social contexts we experience-the towns and cities we live in, our relationships with our parents, peers, teachers; our continuing interactions at work, at home, at play etc
- The term *nurture* is used to describe an *organism's environmental experiences*.

Genes

- The term *nature* is used to describe an *organism's biological inheritance*.

Fertilization

- One begins life as a single celled organism, and develops into a human being made up of trillions of cells.
- According to Poole, Warren, and Nunez (2007) an individual starts to form when the male and female reproductive cells (gametes) combine to form a single cell called a *zygote*. This process is referred to as *conception/fertilization*. This is facilitated by *meiosis*-the process of gamete formation which involves halving the number of chromosomes from 46 to 23. *Mitosis* takes place after the zygote has been formed.
- A zygote has 23 pairs of chromosomes.
- **22** are called *autosomes* and contain most of the genetic information for the new individual.
- The **23rd pair**, the *sex chromosomes* determines the sex.
- Zygotes containing **2 X** chromosomes develop into *females*. □ Those containing **one X** and **one Y** chromosome develop into *males*
- **NB: Sex determination** is influenced by the *male gametes*.
- The nucleus of each human cell contains *chromosomes*.

- Chromosomes are ***threadlike strands of DNA (deoxyribonucleic acid) molecules that carry genetic information.***
- With the exception of the sex cells (sperm and eggs), every cell in humans contains 46 chromosomes.
- These chromosomes operate in 23 pairs, with one chromosome of each pair coming from each parent.
- Each chromosome in turn contains thousands of biochemical ***messengers called genes.***
- Genes are DNA segments that serve as key functional units in hereditary transmission.
- Every person has two genes for each characteristic governed by heredity.
- When genes combine to determine our characteristics, some genes are dominant over others.
- According to the ***dominant-recessive genes principle***, if one gene of a pair is dominant, and one is recessive, the dominant gene exerts its effects, overriding the potential influence of the recessive gene.
- A recessive gene exerts its influence only if both genes of a pair are recessive.
- ***Genotype is a person's genetic heritage***, the actual genetic material.
- ***Phenotype*** is the way an individual's genotype is expressed in ***observable, measurable characteristics***. Phenotype includes physical characteristics (such as height, weight, and eye colour), and psychological characteristics (such as intelligence and personality).

Twins

- ***Identical twins (monozygotic)*** develop from a single fertilized ovum, and are therefore identical in genetic makeup.
- Any differences between them may be due to environmental influences.
- ***Fraternal twins (dizygotic)*** develop when two separate eggs are fertilized simultaneously.
- Fraternal twins are no more alike in genetic makeup than any two siblings born to a pair of parents at different times.

Chromosomal abnormalities and the resultant behaviour

NB:

- Chromosomes exist as pairs so that the proper number (46) is maintained

chromosome trisomy.

- If ***sex determining chromosomes*** are involved in monosomy, this is known ***as sex chromosome monosomy.***
- When other chromosomes are involved in trisomy, this is referred to as ***autosomal trisomy***; and if involved in monosomy, this is known as ***autosomal monosomy.***

Examples of trisomy and monosomy

1. Sex chromosome monosomy

Turner's syndrome: disorder in females in which either an x chromosome is missing, making the person xo instead of xx, or the second x chromosome is partially deleted.

Characteristics: short in stature, webbed neck, may be infertile, poorly developed secondary sex features e.g. breast size, sexual dysfunction in adulthood, broad chest with widely spaced nipples, perceptual and cognitive deficiencies.

2) Sex chromosome trisomy

a)Klinifelter's syndrome: (named after Dr. Harry Klinifelter in 1942): disorder whereby males have an extra x chromosome, making them XXY instead of XY.

Characteristics:-underdeveloped testes, enlarged breasts, reduced body hair, and possible infertility, poor muscular and motor development, speech and language disorders, taller than average, intelligence is within normal limits, though IQ scores may be on the lower end of the scale.

b) Trisomy XYY (Supermale Syndrome)

A sex chromosomal disorder that affects males-characterized by an extra Y chromosome.

Characteristics: heavy beards, unusual tallness, mild mental retardation, particularly in language development, a violent antisocial personality.

c) Trisomy XXX (triple X syndrome)/Super female Syndrome

Individuals with this condition have an extra X chromosome and are female.

Characteristics: delay in speech and language development, lack of coordination, poor academic performance and immature behavior, taller than average, poor muscular and motor development, neuro-maturational delay, timid personality. Sexual development is usually normal.

3)Autosomal monosomy

- Has not been observed in humans because the embryo cannot proceed beyond the early developmental stages.

4)Autosomal trisomy

a)Down syndrome /Trisomy 21: caused by a deviation on the 21st pair of chromosomes; the individual may have 47 chromosomes. It was discovered in 1866 by a British doctor Langdon Down.

Characteristics: small hands, large tongue, severe mental retardation, heart defects, added risk of leukemia, flattened facial features, short stature, poor motor coordination, retarded expressive language, developmental delays in terms of the onset of sitting, standing, walking and speech development.

NB: the probability of giving birth to an infant with Down's syndrome increases with the age of the mother.

b)Trisomy D and E

- Involve an extra chromosome to pairs 13 and 15

Characteristics: deficient forebrain development, small or absent eyes, absence of olfactory tracts.

NB: Trisomy E leads to cardiac problems and kidney defects.

PERCEPTION

Sensation and Perception

- ❖ **Sensation** is the process through which signals from the environment are detected by the sensory receptors, encoded into nervous impulses and then passed on to the brain.
- ❖ Sensation is the process by which the senses gather information about the environment. The sense organs gather information that **makes us consciously aware of our surroundings**. The sensory process and the nervous system make us aware of our surroundings. The basic unit in the nervous system is called the **neuron**. Neurons are found in sensory tissue, along nerve pathways, in the spinal column and in the brain itself. In the sensory tissue there are specialized cells **called receptors**. They receive information from inside the body and from the environment.
- ❖ When sensory information is received by the receptors it goes through the process of transduction. This is the process that **converts physical energy into neural messages** which are electro-chemical in nature. The neural messages then travel along nerve pathways to the peripheral nervous system, then to the brain. The **brain registers what is sensed**.

General characteristics of sensation

- ❖ Sensation begins with a stimulus in the environment. A stimulus is any form of physical energy a light wave, heat, odor, sound, pressure or taste. The stimulus **excites the nervous system**. The stimulus first excites one of the sensory receptor cells. The receptor cells transform the physical energy into neural impulse. This transformation is called transduction. Each sensory system is specialized to respond to a different type of stimulation.

How many senses do Humans have?

Aristotle had identified five senses vision, hearing, smell, taste, touch. Today scientists have recorded ten or more senses. The sensory receptors of all these senses perform the same general function of receiving stimulus from the environment, converting it into neural energy and relaying it to the brain. In the brain the neural impulses are experienced as light, smell, taste, sound etc.

Examples of senses

A) Chemical senses

i) Smell(olfactory system-the sensory system for smell)

- ❖ The receptors for smell are situated in each nasal cavity in a small cell-packed area called the olfactory epithelium.
- ❖ The receptors collect information regarding romatica (odors) to the brain.

ii) Taste(gustatory system-the sensory system for taste)

- ❖ This is closely related to smell.
- ❖ The receptors reside in the tongue in the back of mouth and throat
- ❖ Receptors are in the packs of taste buds
- ❖ They collect chemical information carried in food, and temperature consistency of the food.
- ❖ They can sense four major tastes, sweet, sour, salty and bitter.

B)The skin(tactile)

- ❖ Our skin is the largest sense organ.
- ❖ It protects us from the environment, holds in body fluids, and regulates our internal temperature.
- ❖ It has numerous nerve receptors distributed in varying concentrations throughout its surface.
- ❖ Skin receptors give rise to the sensations of pressure, temperature, and pain.

C)Body senses

i)Kinesthetic sense

- ❖ **Kinesthetic** sense relays information about body position and movement.
- ❖ It relies on sensory receptors located in muscles, tendons and joints. *ii)Vestibular*
- ❖ The **vestibular** sense is concerned about equilibrium.
- ❖ It enables us to catch ourselves when we trip, and helps us to run without falling.
- ❖ It helps to maintain balance. The sensory receptors are in the inner ear, in the three **semi-circular** canals filled with fluid. Each time a movement is made the liquid shifts. *D)Hearing*
- ❖ The hearing sense is the most important in terms of human communication. The energy force called sound comes to us in waves. These waves are **molecules of air travelling** at approx 750 miles an hour.
- ❖ The tone and pitch of sound depend on the number of waves travelling per second. The strength and loudness of the sound is determined by the height of the waves.
- ❖ Of all our senses, hearing is most sensitive in terms of discrimination. We can distinguish many voices, musical notes, friendly and unfriendly greeting.
- ❖ The receptor cells are found in the cochlea which is a bony inner ear structure.

E)The visual sense

- ❖ **Humans are visual animals.** The visual sense is the most important in terms of human perception. About 80% of the information is collected by the eyes. The eye has 70% of the receptors. We think and reason using visual images and symbols.
- ❖ The basis of all visual stimuli is light. Light is an electromagnetic force that travels through space and approx 186, 000 miles per second. Light travels from sources like the sun, candles, and electric bulbs. This energy travels in waves which hit surfaces at different intervals and with different force. The waves of light travel in varying lengths and intensities. All the wave lengths of light appear as colorless white. When these wave lengths strike the surface they scatter and each appears as a different color or hue.

Perception

- Process of selection, organization and interpretation of sensory information/input in addition to the way we respond (meaning we attach to sensation). For perception to occur there must be sensory stimulation to which we attach meaning.

- The brain's process of organizing and interpreting sensory information. Perception is a creation of the brain; it is based on input extracted from sensory organs, such as the eyes, ears, and nose.

Relationship between sensation and perception

- Sensation is the process of **inputting** of information through the senses.
- Perception is the process through which **organization of the information** understanding of what has been sensed occurs (what has been seen, heard, smelled, tasted, and touched).
- Sensation and perception are important in psychology because they are our link with the outside world.
- The information we gather has a great effect on our psychological functioning. Sensation and perception are part of the psychological process by which we gain knowledge about the world. The process of acquiring knowledge is called **cognition** and it starts from sensation and perception which enable learning, memory, thinking and language development.

Importance of perception

- Sensation is a part of the perceptual process. Through perception, we integrate sensations into percepts. A percept is the simplest meaningful pattern. Thus perception helps us generate meaning /make sense of our sensations.
- Our ability to perceive enables us to obtain the information we need to survive.
- We are constantly bombarded by many stimuli that compete for attention. We therefore need to discriminate what is important and what can fulfill our needs.
- The sensory and perceptual process helps us gather information about our environment. This information assists in goal/need attainment.

Perceptual illusions

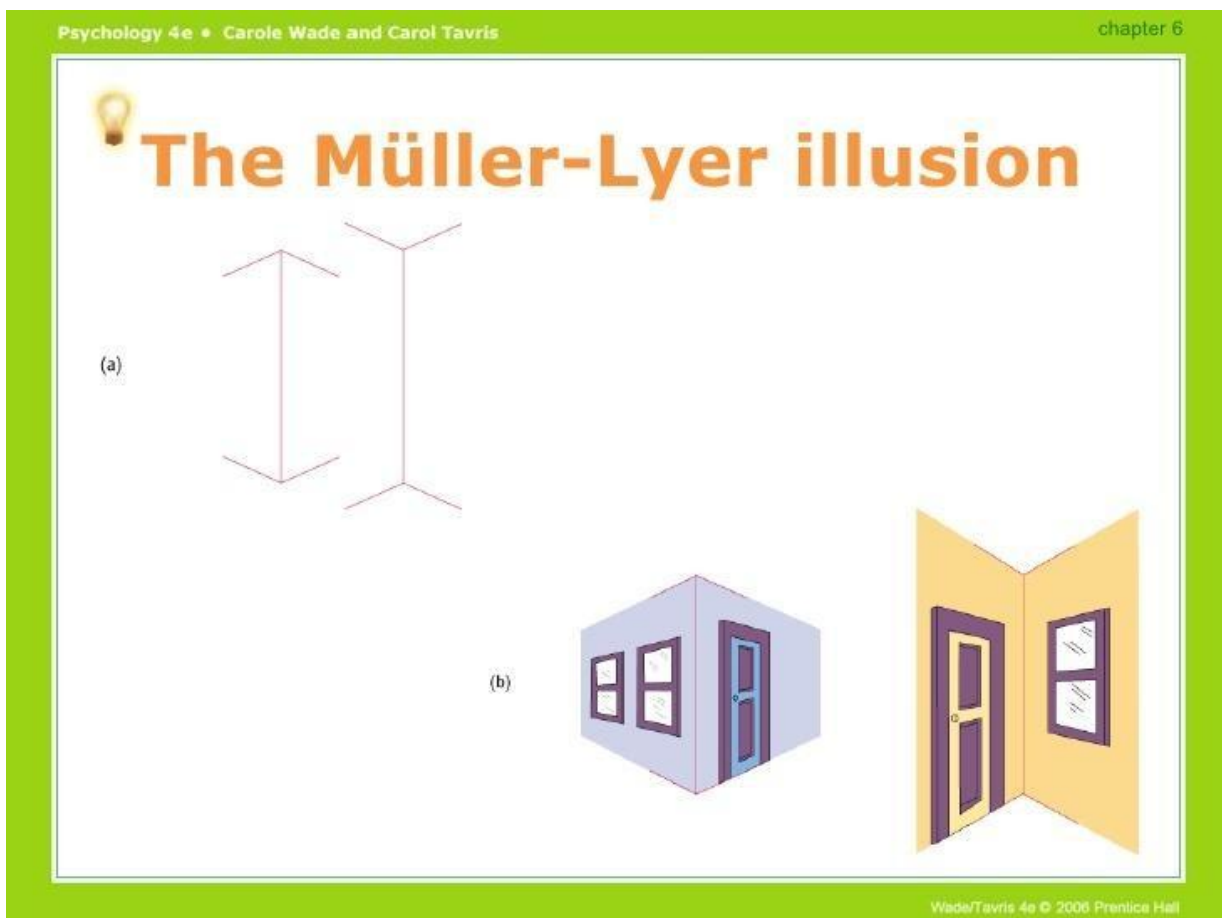
- They demonstrate the ways we use a variety of sensory cues to create perceptual experiences that may or may not correspond to what is out there in the real world.
- They occur when we **misperceive**- seeing things that are not there in reality e.g. Car/train stops but you still feel like the movements are continuing; the moon illusion-the full moon

appears to be much smaller when overhead than when looming over the horizon; when driving at night you think the moon is following you.

- Illusions are incorrect- by understanding how we are fooled into —seeing something that isn't there, psychologists can gain insight into the ways that perceptual processes work.
- Perceptual illusions typically occur because the stimulus contains misleading cues that distort the perceived size or orientation of common objects, giving rise to inaccurate or impossible perceptions.

Examples:

- ***Ponzo illusion***
- ***Müller-lyer illusion***



Looks

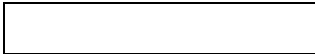
shorter

Looks longer

The top line is perceived as shorter than the bottom, when, in reality, both lines are the same length

- ***Horizontal-vertical illusion:*** in which a vertical line looks longer than a horizontal line even though the two are equal.

❖ ***The poggendorf illusion***



The diagonal line that cuts across the bar(s) is straight, not crooked or jagged.

Thresholds

- Human beings require a certain minimum stimulation of the hearing sense in order to perceive sound. This minimum stimulation is known as the ***absolute threshold***-the smallest amount of stimulus needed in order to notice that the stimulus is present at all. All our sense organs have an individualized absolute threshold.

The absolute thresholds for human beings are quite low. The absolute threshold for:

✚ ***Light***: is a candle light 30 miles away on a dark, clear night

✚ ***For smell***: a drop of perfume diffused throughout a three-room apartment

✚ ***Hearing***: the ticking of a watch from 6 meters(20 feet) in very quiet conditions

✚ ***Taste***: one tea spoon of sugar in two gallons of water (45 litres x 2 = 10 litres of water)

✚ ***Touch***: the wing of a fly falling on your cheek from a height of one centimeter.

- As we sense various stimuli in the environment, we require the ability to detect small differences between these stimuli. For example, a tea taster must be able to detect minute differences in flavor between various qualities of tea. The minimum difference that can be detected with respect to our sensory system is referred to as ***difference threshold or just noticeable difference (jnd)*** -the smallest change in stimulation that you can detect-
e.g. change from no stimulation to stimulation, or change from less stimulation to more. How much stronger must a sound become before you notice that it has grown louder?
- Suppose Kahlo was painting while the sun was setting; at what point would she notice that it had gotten darker in the room, and she should either adjust the way she was painting patterns of light, or call it quits for the day? ***Jnd is the size of the difference in a stimulus property (e.g. brightness of light) needed for the observer to notice a difference*** (Kosslyn and Rosenberg, 2006).

NB: Subliminal perception: stimuli that are too weak for us to notice and yet but are able to affect us (—below threshold) e.g. advert, musical background etc.

Sensory adaptation

- This is defined as the *diminishing sensitivity to unchanging stimuli*. For example, when you walk into a darkened movie theater, you can hardly see anything except the image on the screen. Gradually however, your eyes become *accustomed* to the dark; when you enter a room that is freshly painted initially the odor is strong and uncomfortable but with time one adapts to the odor; mortuary attendants; hospital smells; and noise - initially continuous noise is annoying. But with time the person adapts to it and becomes unaware of it.
- Thus our senses automatically *adjust* to the overall, average level of stimulation in a particular setting.
- When confronted by a great deal of stimulation, they become much less sensitive than when the overall level of stimulation is low.
- Adaptation allows our senses to be keenly attuned to environmental conditions without becoming overloaded.
- When you enter a hushed room, you are able to hear the faintest sound. But when you go out onto a busy street at rush hour, the noise of street traffic would seem deafening, even painful, unless your ears adapted and became less sensitive to noise.

Perceptual organization

- We use structures to categorize various stimuli, resulting in meaningful perceptions.
- These structures are *Figure and Ground Perception* and *Perceptual Grouping*.

a) Figure and Ground Perception

- This is the principle by which we organize the perceptual field into stimuli that stand out (*figure*) and those that are left over (*ground*)-the background, which must be distinguished in order to pick out any figure..
- In order to recognize an object, we need to first perceive this object as being separate from its surroundings. For example- when we write some words on a piece of paper, the words

are perceived as being distinct from the paper. The words assume the characteristics of a figure, while the paper becomes a mere background.

- NB: when figure and ground are similar, the figure is said to be *camouflaged* (animals in the jungle, as well as armies the world over, have long taken this advantage of the perceptual system)

b) Perceptual Grouping

- This requires bringing some order in terms of grouping stimuli together. This ordering is governed by the following principles.

1) Proximity: Things that are near one another seem to belong together- e.g. we see XXX XXX as two groups, and XX XX XX as three groups even though both sets have the same total number of X marks; XOXOXOXOXOXO

XOXOXOXOXOXO

In the above example, we perceive rows, because the principle of proximity creates the impression that the X's and the O's are related due to their nearness to each other.

2) Similarity: Things that look alike tend to be grouped together. So, for example, we see XXXxxx as two groups;

XXXXXXXXXX

OOOOOOOOOO

In this example, similar shapes are grouped together in the perceptual process.

3) Continuity:

- We tend to perceive smooth continuous patterns rather than discontinuous ones. □
Marks/things that fall along a smooth curve or a straight line tend to be grouped together
- Examples:
 - ✚ Eg 1: We see ---- as a single line, not four separate dashes; and we see ---- ---- as two separate lines all eight of the dashes do not fall on the same plane.
 - ✚ Eg 2:

```

OXOOOXXOOOXXOOOXXOOOXXOOOXX
OOXOOXXOOXXOOXXOOXXOOXXOOXXOO
OOOXXOOOXXOOOXXOOOXXOOOXXOO

```

In the above example, we perceive a continuous wave pattern made up of X's, due to the influence of the principle of continuity.

4) Closure:

- People often group elements to create a sense of closure or completeness.
- We tend to close any gaps in a figure, thus enabling one to perceive disconnected parts as being complete.
- So a circle with a small section missing will still be seen as a circle.

5) **Simplicity/Good form:** people tend to group elements that combine to form a good figure. e.g.

This could be viewed as a *complicated 11-sided figure*, but given the preference for simplicity, one is more likely to see it as a rectangle and triangle.

NB: The process of perception therefore requires proper organization of various stimuli.

Perceptual constancies

- Tendency to perceive objects as relatively stable and unchanging despite changing sensory information. Once we have formed a stable perception of an object, we can recognize it from almost any position, at almost any distance, under almost any illumination. A white house looks like a white house by day or by night and from any angle. We see it as the same house.
- Perception of the characteristics of objects (such as their shapes and colours) as the same even though the sensory information striking the eyes changes.
- This enables us to recognize *objects without being deceived by their shape, size, brightness, or color*.
- There are four main perceptual constancies;

a) Shape constancy:

- This is the tendency to perceive familiar objects as having a constant shape in spite of the viewing angle.

Example:

- a dinner plate is viewed as circle, even when it is tilted and the retinal image is oval; a rectangular door will project a rectangular image on the retina only when it is viewed directly from the front. From any other angle, it casts a trapezoidal image on the retina, but is not perceived as having suddenly become a trapezoidal door.
- Stabilization occurs not in the eyes, but in the brain, and is fundamental to the ability to recognize objects and know how to interact with them.

b) Size constancy: This is the tendency to perceive an object as having a constant size in spite of the object being far away from us.

Example:

- We perceive a car/aeroplane as big enough to carry people, even when we see it from afar.

c) Brightness Constancy:

- This is the tendency to perceive objects as having consistent brightness even when their illumination varies.
- Recognition that an object retains the same degree of brightness even though different amounts of light fall on it.

Example:

- White paper reflects 90 percent of the light falling on it, while black paper reflects only 10 percent in well- illuminated conditions. In bright sunlight conditions, the black paper may reflect 100 times more light than does the white paper indoors (McBurney&Collings, 1984), yet it still looks consistently black.
- Regardless of whether you are reading a textbook indoors or outdoors, the white pages and the black print do not look any different to you in terms of the whiteness or blackness.

d)Color Constancy:

- This refers to the tendency to perceive familiar objects as having consistent color even when the illumination changes.
- Hence we tend to see familiar objects as keeping their colours, regardless of information that reaches the eye.

Example:

- A yellow dress looks yellow when observed during the day or evening, when the amount of light diminishes, it looks white, but it is still yellow.
- If you own a red car, you will see it as red whether it is on a brightly lit street or in a dark garage, where the low light may send you a message it is closer to brown or black than red.

Factors that influence human perception

Various factors contribute to perceptual differences such as;

1)Characteristics of the stimulus object

- A stimulus object is anything that draws the attention of our sense organs.
- Objects that are vivid in terms of possessing a sharp distinction between the figure and background are easiest to perceive.
- Colorful objects as well as those with patterns are powerful stimulus objects.
- Conversely, objects that are dull- colored, vague and lacking in contour are easily ignored.

- In class, teachers are encouraged to apply vividness, color and pattern to educational stimuli in order to enhance learning.

2)Culture

- This is a very crucial factor because we are all subject to both cultural and ethnic forces.
- Demands of different cultures lead to greater emphasis on certain senses.
- Eg the Maasai who depend on herding tend to perceive individual differences in what others perceive as a herd of virtually identical animals; wine experts can distinguish subtle differences in the flavor of different vintages that most of us could not detect.
- It has been proposed that Africans perceive round objects better than those with straight lines, while Caucasians prefer the opposite of this.
- In the late sixties, Deregowski's study on Malawians revealed that the group involved perceived colored photographs better than the black and white pictures. The implication of this was that the Malawians lived in a colorful culture, hence their tendency to perceive the colored photographs satisfactorily.
- Turnbull's study on the Pygmies of the Congo (1969) showed that the subjects under study had problems of size constancy. The pygmies were shown some buffaloes at a distance and asked to explain what they saw. They responded that they could see insects. The implication of these results was that pygmies lived in a heavily forested area where trees prevented long distance perception, hence these people did not have ample opportunity to develop size constancy.

3)Context characteristics

- The term 'context characteristics' refers to the setting in which perception occurs.
- This implies that a person may be exposed to a stimulus and the way he/she perceives this stimulus will depend on the circumstances under which it is presented.

Example:

12 13 14 15

- In this example, the number **13** can also be viewed as a **B** depending on the context of application.

4) Selective Attention and Individual Values

- This is the tendency to focus on a specific stimulus and to ignore all the other competing stimuli.
- Usually, the stimulus that captures our attention most is the one that is most vivid, colorful or loud.
- In addition, stimuli containing patterns or emotional potential will enhance selective attention.
- One example we can consider is that of a picnicker and a naturalist in a park. The naturalist will be able to perceive the sounds of birds and animals in the bush as opposed to the picnicker who will generally concentrate on conversation sounds. These two individuals will be in the same environment but will perceive different stimuli due to their special interests, needs and values.

5) Characteristics of the Perceiver

- Individual differences influence the way we appreciate various stimuli. For example, an individual's state of intelligence will influence his/her response to a stimulus.
- One illustration of this is the effect of an examination stimulus. Bright students view examinations as a challenge to prove their proficiency in answering questions, while the dull students look at examinations as a major threat.

6) Our state of mind can also influence the way we perceive.

- For example, a person who is very sick in hospital may not appreciate watching his/her favorite television program. A person who is intoxicated with alcohol will be prone to risky sexual behaviour because of the drink's potential to reduce inhibitions. In this case, risky sexual behaviour is transformed into an acceptable temptation.

7) Motivation

- Our desires and needs shape our perceptions.
- People in need are more likely to perceive something that they think will satisfy their need. E.g. a mirage-people lost in the desert have visual fantasies of an oasis over the next dune.
- Research has found that when people have not eaten for some time, they perceive vague images as being related to food (Morris and Maisto, 2005).

8)Expectations

- Preconceptions about what we are supposed to perceive may also influence perception by causing us to delete, insert, transpose, or otherwise modify what we see.
- This phenomenon of perceptual familiarization reflects a strong tendency to see what we expect to see even if our expectations conflict with external reality.

9)Cognitive style

- As we mature, we develop a cognitive style-a personal way of dealing with the environment-and this also affects how we see the world (Morris and Maisto, 2005).
- Psychologists distinguish between two general approaches that people use in perceiving the world.

✚ **Field dependence:** people who are field dependent tend to perceive the environment as a whole and do not clearly delineate in their minds the shape, colour, size, or other qualities of individual items. If field dependent people are asked to draw a human figure, they generally draw it so that it blends into the background.

✚ **Field independent:** are more likely to perceive the elements of the environment as separate and distinct from one another and to draw each element as standing out from the background.

10)Personality

- A number of researchers have shown that our individual personalities influence our perceptions (Greenwald, 1992).
- For example, normal college students were compared to depressed students in their ability to identify words related to depression.
- Depressed students were faster at identifying adjectives describing personality traits they commonly thought about, such as quiet, withdrawn, hesitant and timid; than they were at identifying adjectives related to traits they rarely thought about, such as extrovert, lively and bold.
- These findings suggest that not only personality, but possibly the presence of a personality disorder, may influence perception.

11) Learning

Everybody is born with innate perceptual tendencies. Learning is that experience which results from interaction with objects and events, people, places. Learning guides the unfolding of innate abilities. This requires that the person be active in the world to be perceived. The world itself must be a rich one with things to see, sounds to hear, objects to manipulate. If the environment is impoverished e.g. a dark room, solitary confinement. The person suffers sensory deprivation which affects perception. Severe sensory **deprivation to the visual sense** causes severe damage to receptor cells in the retina. People who have been blind from birth and who regain sight as adults have difficulty recognizing shapes, faces. They still tend to recognize them by touch. They are unable to learn constancies.

12) Perceptual adaptation or constancy

People perceive the **world as unchanging** even when objects are seen from changing **shapes and sizes**. We perceive the cupness of a cup, the doorness of a door, the goatness of a goat. We perceive constancy of shape, size, colour and brightness. Perceptual constancy relies on experience and knowledge of a particular sensory image before it is modified. It helps us to see the world as relatively unchanging and stable. A coin is round, door is rectangular not trapezium.

13) Subjective experience

People see what they want to see, hear what they want to hear. There is a strong influence of **selective perception** on human behaviours. This is the process of perceiving the environment on the basis of individual needs, values, expectations even if the interpretation is incorrect. People have a way of selecting various pieces of information organizing them in accordance to private motivations depending on one's life experiences, the influence of **cultures, hunger, need for love, loneliness**, need for power all influence perception.

14) Past experience

Past experience keeps the world stable which is positive. It also has a **negative effect** as it keeps people from **expanding their perceptual horizons**. The tendency to perceive the environment in **determined ways is called** and **perceptual set**. The perceptual set influences attitudes, values, interests and other personal judgments. Once a perceptual **set has been formed it is difficult to** change it even when **situations and circumstances change**. The

perceptual set interferes with **relationships between parents and children husbands** and wives, **teachers and students**. The perceptual set is influenced by the people – friends, teachers, relatives, advertisers, governments (brain washing, propaganda). People who want to break from old perceptual sets experience problems. Changes occurs so fast that people have little time to adapt to new perceptual sets – this causes maladjustment.

Extrasensory Perception (ESP)

- According to Santrock (2000) ours eyes, ears, mouth, nose, and skin provide us with sensory information about the external world.
- Our perceptions are based on our interpretation of this sensory information.
- However, some people claim they can perceive the world through something other than normal sensory pathways- this means that they are able to engage in sensation without the intervention to the five senses.
- The field of ESP is nowadays left to experts popularly known as *parapsychologists*, who study the *paranormal*. The term *paranormal refers to phenomena beyond the normal*, as you can deduce from my examples above.
- ESP is *perception that occurs without the use of any known sensory process*.

Parapsychologists have classified various phenomena as follows:

- **Telepathy:** thought transference from one person to another at a distance. Mind to mind communication, where one person sends thoughts to another, who may or may not possess the ability to respond. Telepathy also includes the ability to perceive another's thoughts.
- **Clairvoyance:** Perceiving remote events, that are not in sight, such as sensing that a relative is sick in hospital, how much money is in a sealed envelope, number on a card, a person at a movie theater sensing a burglar breaking into his/her house at that moment etc
- **Precognition:** Perceiving future events-knowing events before they happen. For example, a fortune teller might claim to see into the future and tell you what will happen to you in the coming year; an astrologer might predict that a major earthquake will take place etc

- **Psychokinesis**, which refers to the ability to perform a mental operation to influence a material body or energy system. It is the mind-over the matter phenomenon of being able to move objects without touching them, such as mentally getting a table to rise off the floor or shattering the glass merely by staring at it. A person with this gift can move objects using thoughts.
- **Teleportation**: ability to relocate oneself to places of one's wish.

EPS 100: LEARNING

□ **Behaviorists** define learning —as an *enduring change in behavior potentiality, which occurs as a result of reinforced practice*” (B. Skinner, 1953; Gredler, 2001)

- Focuses on observable behaviors.
- This definition has key concepts the first one being that learning is a **change in behavior**. This key element in the definition informs us that a person who did not know how to **swim and learns** how to swim has experienced a change in behavior from a non-swimmer to a swimmer. Likewise a person who did not know how to use a certain **formula to arrive at answers to mathematics** problems and learns how to use it and now computes the math problems using the formula has learned. That learning is a **relatively permanent change** means that learned behaviors are not easily lost a good example being learning to perform motor tasks like learning how to ride a bicycle or drive a car is never forgotten. Similarly, people learn how to read and write and they never forget. **Learning occurs under conditions of reinforcement** means that people must be motivated by certain incentives to learn. Learning occurs under **conditions of practice means that** for any behavior to be learned a lot of practice is a necessary condition
- **Cognitive psychologists** define learning as —*the mental activity by means of which knowledge, skills, habits, attitudes and ideas are acquired, retained and utilized, resulting in the progressive adaptation and modification of behavior and conduct.*

What behavior changes are not considered as learning?

- Behavior changes that as a result of maturation, organic deterioration, damage to the nervous system are not considered learned because they are not acquired under conditions of reinforced practice and also because they are related to deterioration of adaptive behavior.
- If a behavior is caused by the effect of fatigue, pain or sensory adaptation it is not considered learned because the changes are temporary and the person reverts back to normal behavior once the transitory state shifts.
- Also changes of behavior caused by ingestion of alcohol and other drugs do not fall under the category of learned behaviors because they are temporary states.

Research in Learning

- ❖ Psychology is scientific and therefore explains behavior through empirical observation of data.
- ❖ In order to explain how learning occurs and to develop principles of learning psychologists conduct research. Such research is meant to discover the conditions that facilitate learning and to develop theories.
- ❖ The researchers in this area have used *animals* such as dogs, rats, cats, pigeons, chimpanzees and where research ethics allow human beings have been used.

Why are animals used in learning research?

- ❖ The researchers argue that the humans and animals learn in much the same way and therefore the research findings from animal studies can be generalized to human beings to explain how learning occurs and also to discover the general principles as well as the conditions under which learning occurs.
- ❖ Researchers who use animals in learning research also argue that animals experience and genetic background can be precisely controlled and as such they can find out the extent to which genetic heredity determines learning as well as the environmental conditions that influence what is learned.
- ❖ They also argue that animals' lives are shorter and therefore many generations of animals can be studied.

- ❖ Human beings cannot be subjected to laboratory conditions and studied the way animals are done as it would raise ethical concerns.

Types of learning

The behavioral view of learning

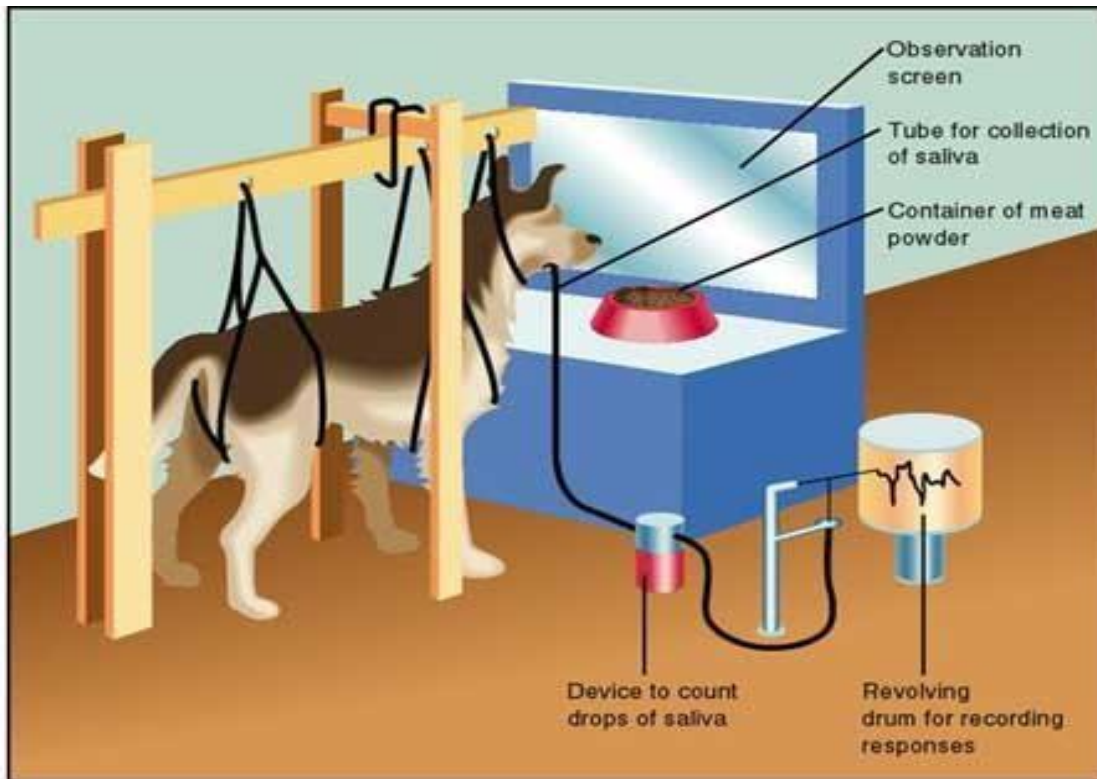
The behavioral approach includes *two views*: namely, *classical conditioning*, and *operant conditioning*.

CONDITIONING

- It involves learning *associations* between events that occur in an organism's environment.

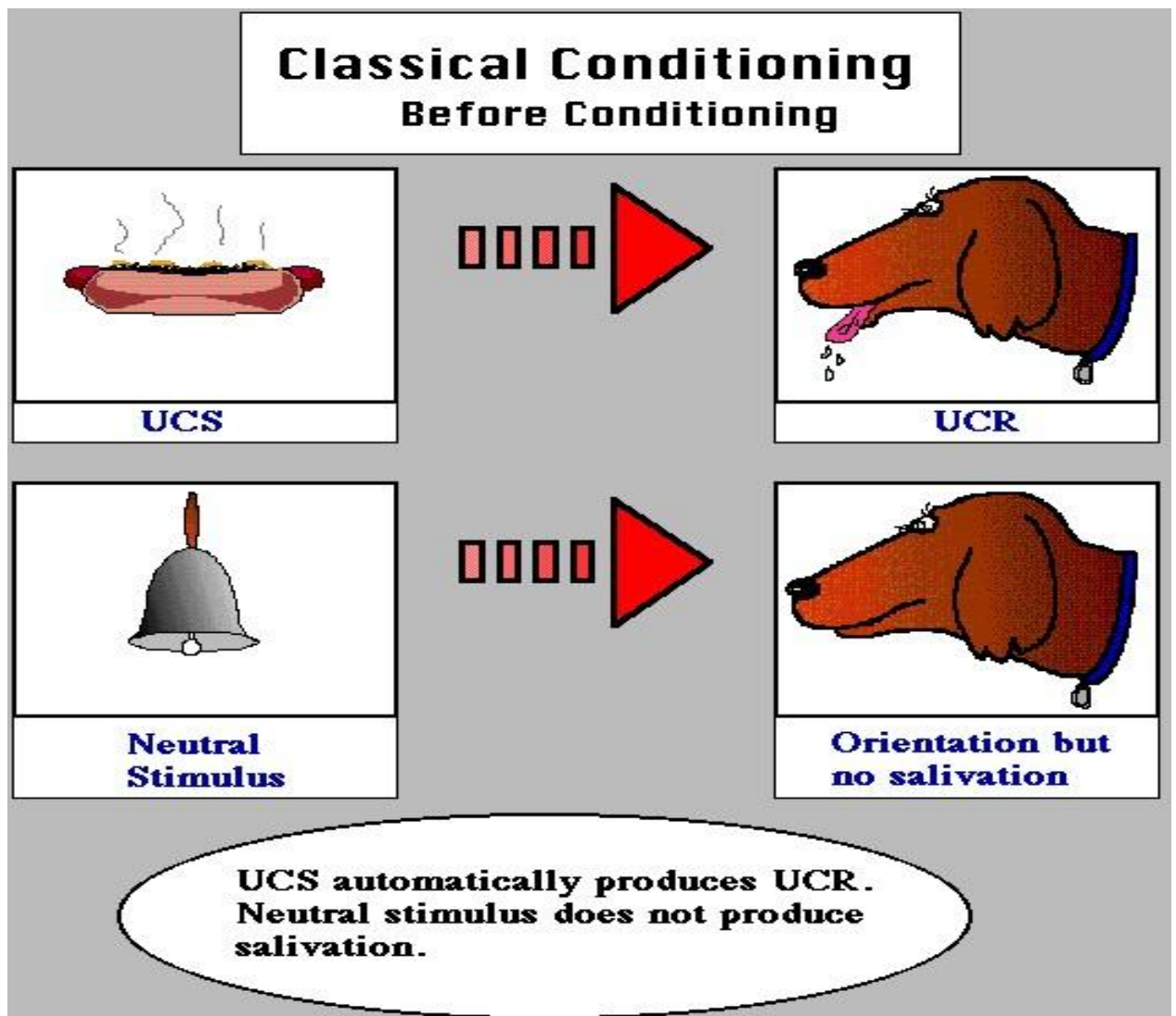
Classical Conditioning

- It grew out of a tradition that can be traced back to *Aristotle (384 – 322 B.C)*, who believed that learning depended on *contiguity – occurrence of events close together over and over again, till they become „associated“* (e.g. lightning and thunder). Later, when only one of these (a stimulus) occurs, the other will be remembered too (response).
- In the early 20th century, the research of *Ivan Pavlov Russian Scientist - (1849 – 1936)* stimulated worldwide scientific research interest in the study of associative learning.
- Experiments in classical conditioning were first conducted by Ivan Pavlov (1849 – 1936).
- Pavlov was a Russian scientist who studied the dogs' digestive system and who received a Nobel Prize for his work in 1903. While working with the dogs in the laboratory Pavlov noted the followings; that when the dogs heard the footsteps of the experimenter or even saw the food bowl it salivated. He argued that this salivation was a learned behavior because only food should cause an organism to salivate.



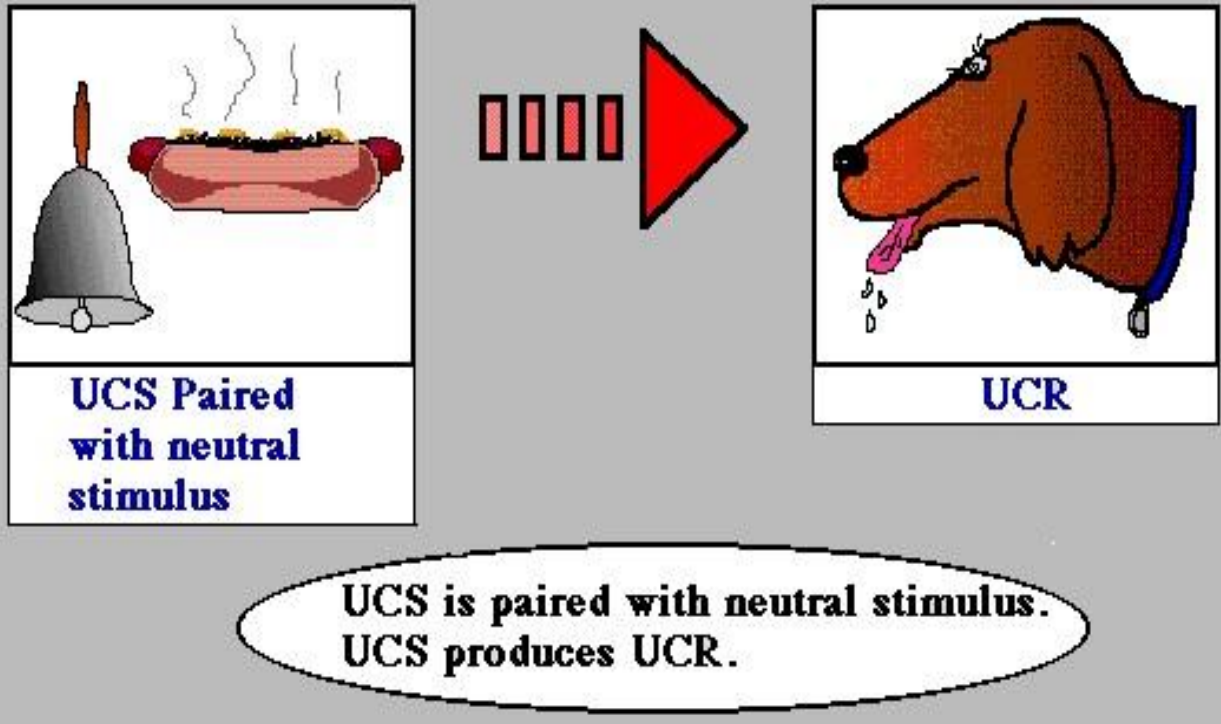
- Contiguity plays a major role in another more complex learning process. It has been called by various names i.e. ***classical conditioning, respondent conditioning, signal learning***.
- ***Conditioning is a psychological process that causes an organism/ person to respond in a particular way to a stimulus*** (anything that elicits a behavior or response) i.e. sights, sounds, smell e.tc. the senses receive from the environment.
- ***Learning through conditioning is therefore similar to learning through association*** i.e. the process where a previously neutral stimulus is able to elicit a certain response.
- ***Pavlov's experiment on a dog*** illustrates how learning takes place through conditioning.
- Using these ***3 elements – food, the salivation and the bell, Pavlov demonstrated that a dog could be conditioned to salivate after hearing the bell.***
- Pavlov began by ringing a bell and recording the dog's response. There was no salivation.
- Then he fed the dog with a piece of meat (S). The response was salivation (R).

- The *food* in this case was an *unconditional stimulus (US)* since it brought forth an automatic response of salivation. The *salivation* was an *Unconditioned Response (UR)*, again because it occurred *naturally*/ automatically.
- He noted that a neutral stimulus (such as a bell) at first prompted no response.

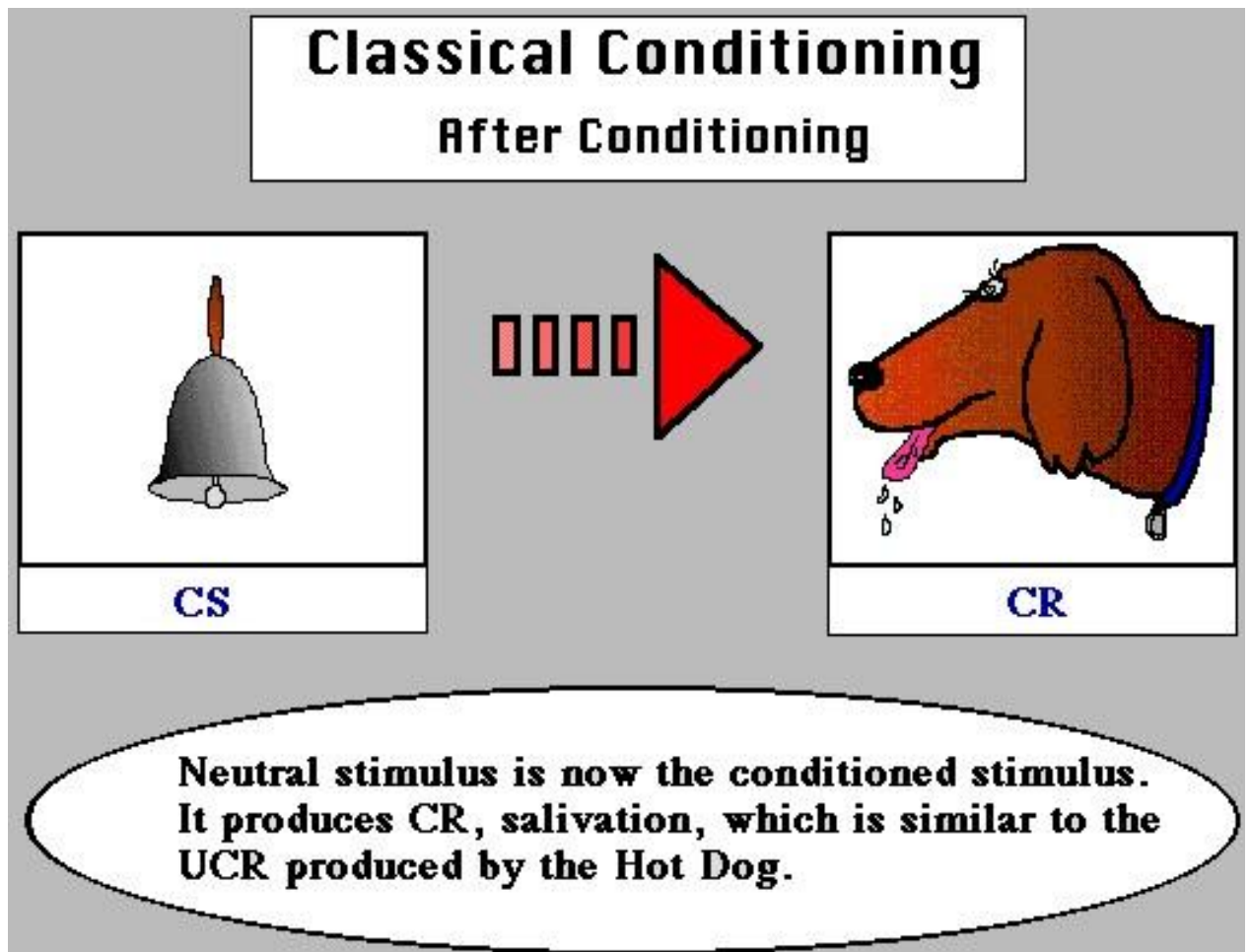


Classical Conditioning

During Conditioning



- Pavlov's experiment showed that if a *previously neutral stimulus* is *paired* with an *unconditioned stimulus*, the *neutral stimulus becomes a conditioned stimulus*.
- He did this by *contiguous pairing* of the *sound/tone with food*.
- He rang the bell and then quickly fed the dog. After repeating this several times, the dog began to salivate after hearing the sound, but before receiving the food.



- The response of salivating after the bell, now a CR was very similar to the original response to the food.
- The *bell had become a CS* that could *bring forth salivation by itself*.
- What Pavlov demonstrated was how *stimulus-response associations* –the *basic building blocks of learning* –are formed by events in an organism's environment.

Basic processes in classical conditioning

- *Acquisition*–the formation of a new conditioned response tendency
- Stimulus *contiguity-association* between two events: Learning process must be done in such a way that that it is possible to associate unconditional responses.
Pavlov thought that **the** key to Classical conditioning is the pairing of stimuli.

- The process must be ***repeated severally*** until the association is formed fully.
- The association must be ***reinforced*** by use of reward or punishment. **Principles of**

Classical Conditioning

1) The principle of reinforcement

When the dog is put in the experimental situation and the tone is sounded and then food is given, this constitutes reinforcement. ***The food is a pleasant event presented to the dog and it constitutes positive reinforcement.*** Content should be sequenced to ensure high correlation and integration of all learning tasks – continuity in curriculum content.

Learning of skills and concepts is easily reinforced.

2) The principle of extinction

Extinction refers to the ***dying of the learned response***, which occurs when ***reinforcement is withdrawn***. It can also be described as the ***gradual weakening and disappearance of a conditioned response tendency***. This occurs when a dog has been conditioned to associate the tone or bell with presentation of food and after some time the food is not delivered. When the experimenter rings the bell or sounds the tone without giving the dog food, the learnt behavior disappears.

3) The principle of spontaneous recovery

Spontaneous recovery is the ***reappearance of a learnt response without the reintroduction or reinforcement***. It occurs if a dog that was initially conditioned to associate the sound of the bell or tone with the presentation of the food is taken through the process of extinction. It is then given some resting time where no bells are heard. If this dog is returned to the experimental situation again and the bell is sounded, it will salivate even if no food is given. However, this response dies very fast if reinforcement is not reintroduced.

4) The principle of generalization

Generalization refers to ***responding to stimuli, which is similar to the original stimuli as long as both are reinforced***. This occurs when the experimenter reinforces the dog with food every time he sounds different types or tones or bell sounds. In this training, the dog learns that every time there is a bell sound, whatever the type, there will be food.

Therefore, the *dog learns to respond to similarities. So when there is a tone similar to the one it was conditioned to, it salivates.*

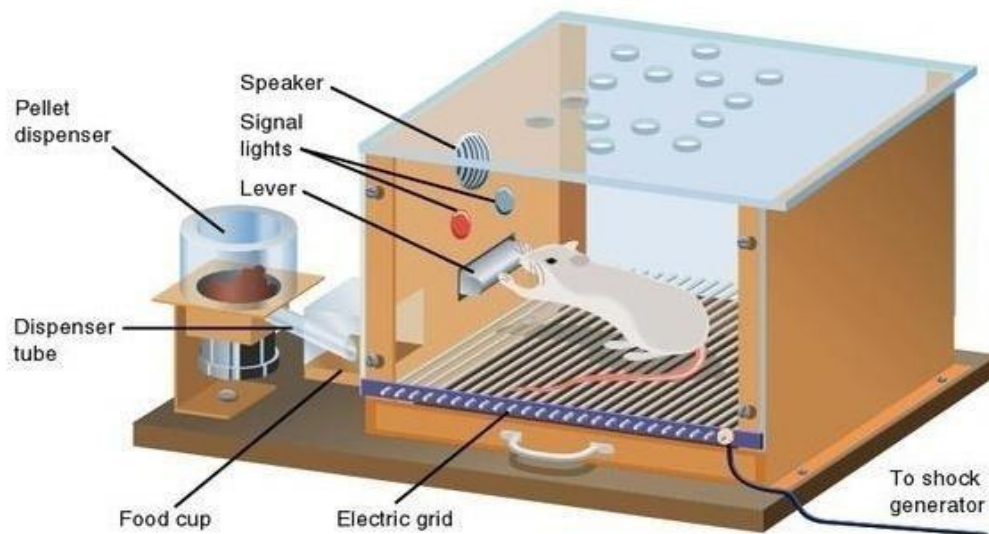
5)The principle of discrimination

Discrimination refers to learning *to pick out the differences in the stimuli* and therefore *responding to a very specific stimulus*. It occurs when an organism that has learned a response to a specific stimulus does not respond in the same way to new stimuli that are similar to the original stimuli. This occurs when a dog has been conditioned to respond to a *particular stimulus*. If the experimenter introduces other stimuli, he does not accompany them with the reinforcement. *As a result, the dog learns to pick out the differences in stimuli and to respond to very specific ones*. Discrimination training takes a much longer time and more trial to establish.

Operant/ instrumental conditioning

- It is a form of learning in which voluntary responses come to be controlled by their *consequences*.
- It is associated with ***B.F Skinner (1904 – 1990)***
- Skinner (1953), used boxes, known as Skinner boxes to study learning.
- The box had a level/ bar at one corner, which was connected to a food magazine containing food pellets.
- A hungry rat would be put in the Skinner box. Hunger would motivate the rat to move. Each movement was called a trial.
- The rat would accidentally touch the bar and operate the food magazine, which would release the food pellets.
- Eventually, the rat learnt to associate the bar- pressing behavior with food.

□ Therefore, the bar- pressing behavior was the learnt response which was accompanied by the food which was the reinforcement event.



Reinforcement contingencies in Operant Conditioning

Skinner also studied the different kinds of relationships between behaviors and their consequences, which he termed as behavioral contingencies i.e.

1) **Positive reinforcement:** Process of *increasing the frequency or duration of a behavior* as the result of *presenting a reinforcer/ rewarding stimulus* e.g. a teacher's positive comments or compliments that encourage students. Good grades, tasty meals, scholarships, promotions, attention and flattery are all positive reinforcers.

2) **Negative reinforcement:** The *frequency of a response increases* because it is followed by the *removal of an aversive (unpleasant) stimulus*. It involves a favorable outcome that strengthens a response tendency. e.g.

- A father nags at his son to do homework. He keeps nagging. Finally, the son gets tired of having to hear the nagging and does his homework. The son's response (doing homework) removed the unpleasant stimulus (nagging).
- A student chronically misbehaves in class and is sent to the Dean of Students.

Next day, the misbehavior occurs sooner and you send the student out again. The intent was to stop misbehavior; however, it was negatively reinforced.

3) Punishment: presenting *aversive/ unpleasant stimuli to decrease undesirable behavior*. It occurs when an event following a response weakens the tendency to make that response. Punishment typically involves presentation of an aversive stimulus (e.g. spanking). However, punishment may also involve removal of a rewarding stimulus
(e.g. taking away a child's TV-watching privileges).

4) Primary and secondary reinforcers

Primary

They are events that are *inherently reinforcing* because they satisfy biological needs eg water, pain avoidance, and temperature regulation.

Physiological needs are *unlearned and survival related*

Secondary

They are *learned or acquired*. They are not related to survival. The person acquires these needs as he/ she interacts with other people. For example, good grades, praise, attention, approval, success, flattery, applause, and affection.

Schedules of reinforcement

- A schedule of reinforcement is a specific pattern of presentation of reinforcers over time.
- There are *two schedules* of reinforcement:
 - ***Continuous:*** *Every instance* of a desired behavior/designated response *is reinforced*.
 - ***Partial:*** *Reinforcing* a response *only part of the time*. Skinner (1953) developed partial reinforcement timetables that determine when a response will be reinforced.

4 Kinds of Partial Reinforcement schedules according to B.F. Skinner i) Fixed Ratio (FR):-Predictable reinforcement

-A reinforcer is given after a ***fixed number of desired responses***, regardless the amount of time it takes e.g. praising the student after every fourth correct response.

ii) Variable Ratio (VR):-Unpredictable reinforcement

-It is one in which the ***number of behaviours required for reinforcement is unpredictable***, although it is certain that the behaviours will eventually be reinforced- e.g. reinforcing the 10th correct response, then 5th, then 14th etc.

iii) Fixed Interval (FI): -Predictable reinforcement

-Based on ***set time intervals*** e.g. giving students a final exam or giving a weekly test.

iv) Variable Interval (VI): -Unpredictable reinforcement

-It is based on ***variable time intervals*** e.g. giving surprise quizzes. The VI is effective for maintaining a high rate of behaviour, while being highly resistant to extinction.

INTELLIGENCE

Definition

❖ According to McDevitt and Ormrod(2002) theorists define intelligence in many different ways, but most agree it has several distinctive qualities:

- ✚ It is ***adaptive***, such that it can be used flexibly to respond to a variety of situations and problems.
- ✚ It involves ***learning ability***: People who are intelligent in particular domains learn new information and behaviours more quickly and easily than people who are less intelligent in those domains.
- ✚ It involves the use of ***prior knowledge*** to analyze and understand new situations effectively.
- ✚ It involves the complex ***interaction*** and ***coordination of many different mental processes***.

✚ It is ***culture-specific***: what is intelligent behaviour in one culture is not necessarily intelligent behaviour in another culture.

- ❖ Intelligence is therefore the ability to benefit from experiences and therefore modify future behaviours to accomplish new tasks successfully.
- ❖ Intelligence is also the ability to abstract or adjust to one's environment. This involves the ability to learn, solve problems, memory, reasoning etc.

Components of Intelligence

1. Ability to reason/Ability to think abstractly
 2. Ability to learn/learn from experience
 3. Ability to adapt to environment
 4. Ability to solve problems
 5. Understand and learn complex material
 6. Mental quickness
- Because of its general nature, intelligence integrates cognitive functions such as perception, attention, memory, language, or planning.
 - On the basis of this definition, intelligence can be reliably measured by standardized tests with obtained scores predicting several broad social outcomes such as educational achievement, job performance, health, and longevity.
 - To allow for comparison of test scores among persons, ***L. Terman devised the concept of intelligence quotient (IQ):***

$$IQ = (MA/CA) \times 100$$

MA = mental age

CA = chronological age

$$IQ = \frac{MA}{CA} \times 100$$

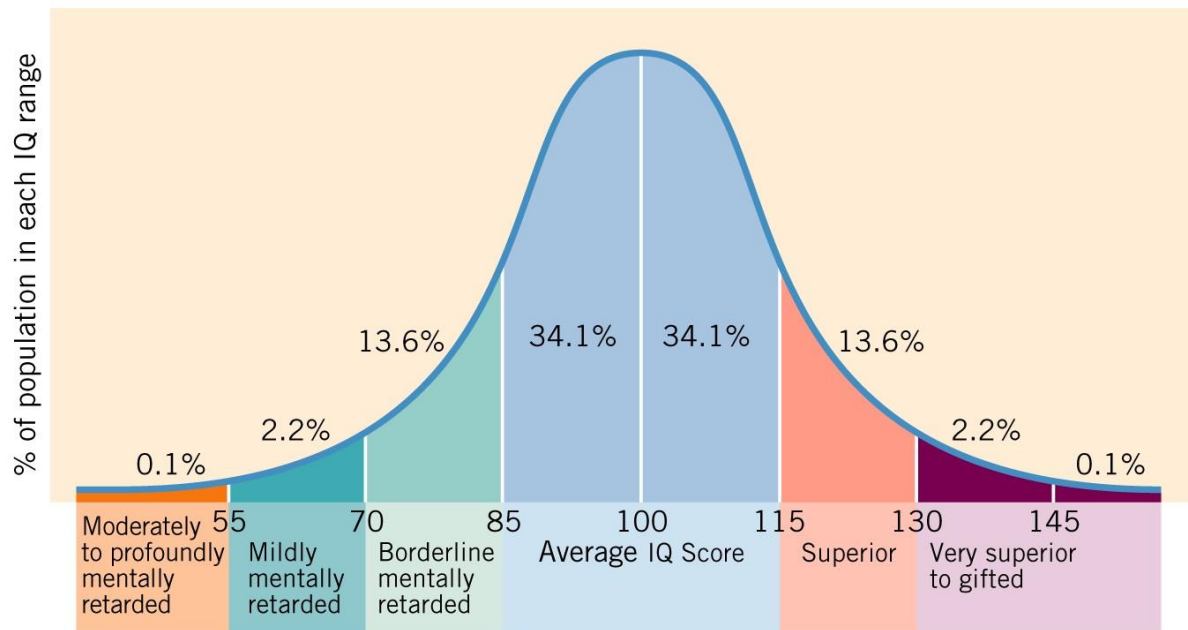
- 1) Chronological Age (CA)-This is the age from birth.
- 2) Mental Age (MA)- This is the level of mental development determined by standardized intellectual mental tests for various age groups. If a child has a CA=3 and MA=4, he is capable of doing work of an older child (4 years) so deemed bright.
- 3) IQ (Intelligent Quotient)-This is the rate of mental development or speed calculated as follows;

Using the example in number 3) the child's IQ would be

$$IQ = \frac{4}{3} \times 100 \quad \text{which is about 133 so very bright}$$

Classification of individuals based on IQ

	IQ	Classification
	150+	Genius(deviates from normal curve)
	131-149	Very superior
	115-130	Superior
	85-114	Normal/ average
	70-84	Dull
	50-69	Mentally retarded (moron) MA 8-12
	20-49	Severely mentally retarded(imbecile) MA 3-7
	Less than 20	Profoundly MR(Idiots)MA less than 3 uneducable



Mental retardation

- Children with mental retardation show developmental delays in most aspects of their academic and social functioning.
- They exhibit below-average intelligence and deficits in adaptive behaviour

Causes of mental retardation

- Birth injuries leading to brain damage
- Prolonged labour (difficult birth) leading oxygen deprivation i.e. anoxia hence brain damage
- Inherited deficiencies/Chromosomal deficiencies-Down syndrome
- Inbreeding (marrying relatives who share genes e.g. most royal families produce feeble minds, no intellectuals)
- Ingestion of toxic substances by mother when expectant e.g. alcohol, drugs, eating soil which could have lead.
- Malnutrition especially lack of proteins leading to fewer brain cells
- Age of mother – past 40 years, children of low intelligence especially if it is the first child.

- Diseases such as venereal diseases e.g. syphilis affects baby's nervous system, German measles
- Radiation that is to say X rays
- Parental neglect or an extremely impoverished and unstimulating home life
- Iodine deficiency leads to cretins i.e. short babies with big heads and low IQ because glandular system is affected
- RH factor due to incompatibilities of blood between couples. The antibodies from mother's blood fight the perceived stranger/enemy in the womb i.e. the foetus antigens that carries father's blood cells leading to either miscarriage, still birth or mental retardation. Only about 2% of mothers have negative rhesus factor. **Structure of intelligence**

It consists of:

- 1) **Abstract intelligence**-ability to deal with formulas, symbols in the brain e.g. statistics, calculations without anyone even noticing
- 2) **Concrete intelligence**-ability to deal with objects mechanically e.g. plumbers, mechanics, carpenters without necessarily understanding the reasons behind what they do. Such people are just practical as opposed to theoretical people.
- 3) **Social intelligence**-ability to deal with social situations and people appropriately and effectively e.g. not making irrelevant jokes, wrong talk, embarrass others- not being a social idiot
- 4) **Emotional intelligence**-ability to identify, assess or control the emotions of oneself, others or groups. This can be self-perceived or perceived by others as they look at how one deals with people and situations.
- 5) **Practical intelligence**- knowing what to do in situations and going ahead and doing it (implementers) e.g. going back to school in order to get promotions in future.

Theoretical perspectives of intelligence

1) Charles Spearman (British psychologist)

- Used a psychometric approach-a theoretical perspective that portrays intelligence as a trait (or set of traits) on which individuals differ.
- **Spearman** noted positive correlations to several different cognitive tests. He speculated that there must be a general mental factor that explained the consistent performance – Intelligence **g**-factor .
- He also noted that there are certain inconsistencies regarding performance in certain areas. He speculated that there were specialized traits that denoted this occurrence intelligence **s**-factor.
- I.e.

✚ **g**-generalized ability, contributes to all intellectual activities/used in a wide variety of tasks and content domains

✚ **s**-specialized abilities- a number of narrow abilities involved in executing particular tasks

2) Raymond Cattell and John Horn

❖ They came up with two distinct aspects of intelligence:

i) *Fluid intelligence*

- Largely the result of inherited biological factors.
- The ability to perceive relationships and solve relational problems of the type that are not taught and are relatively free of cultural influences.
- It involves the ability to acquire knowledge quickly and thereby adapt readily to new situations.
- It peaks in late adolescence and begins to decline in the early 20s.
- It relates more to novel tasks, especially those that require rapid decision making and are largely nonverbal in nature.

ii) *Crystallized intelligence*

- It depends on both fluid intelligence and experience and so it is influenced by both heredity and environment.

- It continues to increase throughout childhood, adolescence, and most of adulthood.
- The ability to understand relations or solve problems that depend on knowledge acquired from schooling and other cultural influences.
- It is more important for familiar tasks, especially those that are heavily dependent on language and prior knowledge

3) Sternberg's Triarchic Model

Spear Sternberg proposed that intelligence is comprised of 3 aspects:

a) **Analytical intelligence**-refers to mental processes e.g. ability to learn how to do things, acquire new knowledge, solve problems, and carry out tasks effectively. Most intelligence tests assess analytical intelligence.

b) **Creative intelligence**: ability to adjust to new tasks, use new concepts, combine information in novel ways, respond effectively in new situations, gain insight, and adapt creatively.

c) **Practical intelligence**: people who score high are good at finding a multitude of solutions to practical and personal problems. They make the most of their talents by seeking out situations that match their skills, shaping those situations so they can make optimal use of their skills, and knowing when to change situations to better fit their talents.

4)Thurstone's Multiple intelligences theory/ multifactor theory of intelligence

- 1) Number ability-being able to perform simple arithmetic problems rapidly and without errors
- 2) Word fluency-ability to produce words rapidly and appropriately
- 3) Verbal factor-ability to find similarities and differences or contrasts in the meaning of words
- 4) Memory-ability to memorize rapidly and accurately by rote
- 5) Reasoning-ability to find underlying concepts
- 6) Spatial relations-ability to identify spatial relations in pictured objects(what can fit where and perform pictorial representations)
- 7) Perceptual speed-ability to recognize likeness and differences in visual forms rapidly and accurately

6)Howard Gardner's Eight Intelligences

Type of intelligence	Description	Examples of relevant behaviour
Linguistic	Word smart-ability to use language well	✓ Making persuasive arguments ✓ Writing poetry ✓ Identifying subtle nuances in word meanings
Logical mathematical	Number-reasoning smart-ability to manipulate abstract symbols	✓ Solving mathematical problems quickly ✓ Generating mathematical proofs ✓ Formulating and testing hypothesis about observed phenomena
Spatial intelligence	Picture smart-ability to reason well about spatial relations	✓ Conjuring up mental images in one's mind ✓ Drawing a visual likeness of an object ✓ Making fine discriminations among very similar objects
Body kinesthetic	Body smart-ability to plan & understand sequence of movement	✓ Dancing ✓ Playing basketball
Musical	Music smart-ability to compose, appreciate and understand music	✓ Playing a musical instrument ✓ Composing a musical work ✓ Having a keen awareness of the underlying structure of music
Interpersonal	People smart-ability to understand others & social interactions/notice subtle aspects of other people's behaviours	✓ Reading another's mood ✓ Detecting another's underlying intentions and desires ✓ Using knowledge of others to influence their thoughts and behaviours

Intrapersonal	Self smart-ability to understand yourself awareness of one's own motives, feelings and desires	✓ Discriminating among such similar emotions as sadness and regret ✓ Identifying the motives guiding one's own behaviour ✓ Using self-knowledge to relate more effectively with others
Naturalist	Nature smart-ability to observe aspects of the natural environment well	✓ Identifying members of various species ✓ Classifying natural forms e.g. rocks, types of mountains ✓ Applying one's knowledge of nature in such activities as farming, landscaping, hunting etc

Measurement of Intelligence

- Intelligence testing is important to help in designing of the right educational/academic programmes/curriculums and consequently matching them to the job market.
- It also helps educators know where children fit in the educational programmes

Disadvantages of group IQ tests

- Group tests tend to emphasize speed, verbal factors and reading comprehension to a greater extent than do individual tests
- Motivation and fatigue not noticeable
- An uninterested, anxious, confused child may get a poor than is typical of his performance.

Intelligence tests

They include:

i) Weschler Intelligence Scale for Children-WISC

- ✚ WISC –III: is designed for children and adolescents aged 6-16
- ✚ It contains 13 sub-tests, each involving either verbal responses or object manipulation.

ii) Stanford-Binet Intelligence Scale

- ✚ Can be used with people ages 2 through adulthood

- ✚ The individual being assessed is asked to perform a wide variety of tasks, mostly involving verbal material and response(e.g. defining vocabulary words, finding logical inconsistencies in a story, or interpreting proverbs) but a few involving manipulation of concrete objects(e.g. cardboard boxes, buttons or blocks).
- ✚ It yields an overall IQ score, plus more specific scores in verbal reasoning, abstract/visual reasoning, quantitative reasoning, and short-term (i.e. working) memory.

iii) Universal Nonverbal Intelligence Test (UNIT)

- ✚ Designed for children and adolescents ages 5-17
- ✚ The UNIT consists of six sub-sets involving memory or reasoning regarding visual stimuli.
- ✚ Its contents was chosen from objects and symbols presumed to be universal across all cultures
- ✚ Instructions are given entirely through gestures, pantomime, and modeling and the child responds by either pointing or manipulating objects.
- ✚ Ideal for children who have hearing impairments, children who have languagerelated learning disabilities, and children for whom English is a second language

Problems with intelligence tests

- They are not culture-fair (items used/asked about are not within the immediate environment of test takers)
- They fail to get a native inborn ability
- A child's performance and resulting IQ are greatly affected by his home background and other social factors.
- The tests leave untouched many important aspects of personality such as interests, motives, attitudes, social adaptability etc
- Group tests do not allow for clarifications
- Test takers may not be motivated enough

Example-Which of the following is taken before the main dish?-ice cream, coffee, soup, tea.

The answer depends on one's culture as well as socio-economic factors.

To resolve this, tests that were appropriately normed (*culture-fair tests*) were introduced that take into consideration the kinds of experiences that the children have had.

NB: There cannot be culture-free tests since all tests are developed in a culture (test developers and takers are from a certain culture)

Conditions that must be fulfilled for a test to be considered culture-fair

- Test items must be familiar/similar to those in the environment(dont tell Nairobi/urban children to classify wild animals since they only see kittens, dogs, horses)
- Language of the test must be well understood by test takers
- Test takers must be well motivated to take it otherwise the differences in the scores may not be easy to interpret-is it due to IQ, boredom, demotivation etc

Determinants of Intelligence/ factors affecting intelligence

Evidence for hereditary influences

1) *Heredity (role of genes)*

- ❖ Studies done show that if you separate siblings and bring them up in different environments, then check their IQ, there is a high correlation between IQ levels of family members. The closer the blood relationship, the greater the correlation.
- ❖ ***Twin studies*** with identical twins have produced results that show the role of heredity. There is a high degree of concordance when identical twins are reared together (almost a positive perfect correlation 0.8). If not reared together, the environment plays a role and reduces correlation. In a review of twin studies,Bouchard and McGue (1981) found these average (median)correlations:

	Correlations of twin's IQs
Identical twins raised in the same home	.86
Identical twins raised in different homes	.72
Identical twins raised in the same home	.60

MacDevitt and Ormrod(2002) concur that the correlation of .72 indicates that identical twins raised in different environments tend to have very similar IQ scores. These twins are more similar to each other than are fraternal twins raised in the same home.

- ❖ ***Adoption studies:*** Adopted children share a similar environment with their adoptive parents; they share a similar genetic make-up with their biological parents. When researchers obtain IQ scores for adopted children, and for both their biological and adoptive parents, they typically find that the children's IQ scores are more highly correlated with the scores of their biological parents than with the scores of their adoptive parents (Low IQ correlations reported between adopted /foster children and foster parents).
- ❖ Advocates of heredity include Sir Francis Galton, and Arthur Jensen. Galton looked at the top scientists in Britain and found out that they came from families where parents were intellectually great. He therefore concluded that genius is inherited and recommended for selective breeding (geniuses marrying geniuses) to improve the human race
- ❖ Kallikak had an affair with a mentally retarded girl and they gave birth to a mentally retarded child. He later married a normal woman and their children were not retarded. He therefore concluded that mental retardation is inherited (sample too small for generalization, there could have been other causes not studied).
- ❖ Jensen mated fast learning rats with fast learning ones and slow with slow ones. The offspring of fast learning ones learnt to go through a blind maze faster than those of slow learning rats. He also gave tests to both white and black children and found out that black children scored consistently low grades than the whites. He therefore concluded that in the development of intelligent, heredity outweighed the environment (socio-economic factors??)
- ❖ Measures of information processing speed correlate with IQ scores.Speed of processing depends on neurological efficiency and maturation,which are genetically controlled.

2) Evidence of environmental influences

- ❖ Studies of identical twins brought up under different environments (stimulating vs impoverished one) - their IQ, height, weight is measured (ideally if living together all these would be same since they are identical!). When brought up separately the correlations between IQ was 0.4, height 0.8, weight 0.67, scholastic aptitude that depends on IQ was 0.4. Heredity affected only height but the rest were determined by the environment. If heredity was that powerful, it could have defeated environment and produced high correlations for all measures.
- ❖ **Sex**-some studies have reported that boys are superior to girls in mathematical, scientific and mechanical aspects though this could also be an aspect of gender socialization/cultural conditioning hence attitudes rather than biological differences. Girls are reported to be better in verbal task.
- ❖ **Race?**-no evidence of differences in IQ due to race
- ❖ **Stimulating home environment:** Stimulating home environments –those in which parents frequently interact with their children, make numerous learning and reading materials available, encourage the development of new skills, use complex linguistic structures in conversation-and are associated with higher IQ scores in children (heredity may produce intelligent children, but more intelligent parents provide more stimulating environments).
- ❖ **Effects of early nutrition:** Severe malnutrition, either before birth or during the early years of life, can limit neurological development and thereby have a longterm influence on cognitive development. Poor nutrition in the early years is associated with lower IQ scores, poor attention and memory, and lower school achievement.

MOTIVATION

Definition of terms

- **Motivation**- the process that *initiates directs and sustains behaviour* to satisfy physiological or psychological needs or wants.

- Also defined as the *driving force or impetus behind behavior, thoughts and actions*.

Motives: Needs or desires that energize and direct behaviour toward a goal.

- *Inborn motives* are those that are acquired naturally beginning in early childhood, for example, the need for food, water, shelter and oxygen.
- *Acquired motives* are those that are developed through the interaction of an individual with a given environment. For example, the need for social approval is learnt from interacting with other people.

Homeostasis- The tendency of the body to *maintain a balanced internal state* with regard to oxygen level, body temperature, blood sugar, water balance and so forth. When this state is disturbed, a drive is created to restore the balance.

Drive- A state of tension or arousal brought about by an underlying need. It motivates an organism to engage goal-directed behaviour.

- Primary drives-** these are unlearned and serve to satisfy biological/physiological needs e.g. hunger, thirst, sex, sleep, exercise etc. They must be satisfied.
- Secondary drives-** these are acquired needs e.g. need for recognition, achievement, self-esteem, belongingness e.t.c.

Incentive: An external stimulus that motivates behaviour e.g. money, fame etc.

Primary characteristics of motivation

- Activation-** the initiation of motivated behaviour. It involves taking the first steps required to achieve a goal.
- Persistence-** the faithful and continued effort put forth in order to achieve a goal.
- Intensity-** the focused energy and attention applied in order to achieve a goal.

Types of motivation

a) Intrinsic Motivation- the desire to perform an act because it is satisfying or pleasurable in and of itself.

E.G.

- ✚ A student working hard in a subject because he/she finds it interesting.
- ✚ A student reads several books a week because reading is fun.
- ✚ A person donates large sums of money to a university to fund scholarships for deserving students.

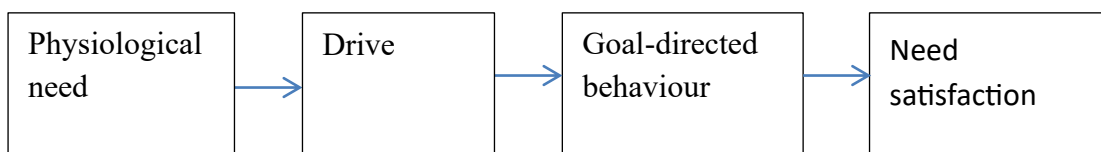
b) Extrinsic motivation- the desire to perform an act to gain a reward or to avoid an undesirable consequence. E.G.

- ✚ A person agrees to donate huge sums of money to the university to construct a building, provided it will bear the family name.
- ✚ A student reads two books a week to avoid losing TV privileges.
- ✚ A student studying to avoid being kicked out of school.

Characteristics of Motivated Behaviour

- The source of motivation is the *need that has to be satisfied for an organism to achieve internal equilibrium*. The satisfaction of a need constitutes reinforcement because this action is rewarding in itself.
- When a *goal is attained, the motivated behaviour is terminated*. For example, after we sit the final examination, we are not likely to continue studying since the goal has been achieved.
- *Motives are internalized*; hence we can only infer them from our behaviour. This means that we are not able to tell accurately what motive a person has by just looking at their behaviour or facial expression.

The Motivation Cycle



- When equilibrium is disturbed, a drive is created to restore balance. The drive motivates an organism to engage in behaviour that will satisfy the need, and reduce the tension thus restoring equilibrium/balance.

Types of motives

a)Primary Motives

- ❖ Primary motives are also referred to as primary needs, physiological needs, biological needs and tissue needs. They originate from within an organism and are influenced by the biological requirements of an organism.
- ❖ Primary needs *motivate the behaviour of an organism in directions that lead to their satisfaction*. For example, a hungry animal is motivated to look for food, which is necessary for survival. Prolonged failure to satisfy some of the primary needs can result in impaired health, or death.
- ❖ All *primary needs are inborn*, meaning that they arise from the organisms inherited structures. Primary needs include the need for water, food, oxygen, rest, asleep, shelter, warmth when cold, coolness when hot, the need to relieve the sex drive, pain and bowel (and bladder) tensions.

NB: Primary needs cannot be satisfied permanently. For example, when we have a meal, the state of satisfaction is temporary, since the hunger drive will be renewed in a short while.

b)Secondary Motives

- ❖ Secondary motives are those drives that an individual learns through interaction with agents in the environment. They are also referred to as learned motives, acquired motives and psychological needs. They are not basic to the survival of an organism. Examples include the need for fame, prestige, power, material possessions, money, greater knowledge, self-esteem and social approval.
- ❖ Secondary needs are important to man because they involve *interpersonal relations with other people*. These needs are frequently overwhelmed by the physiological needs as well as obstacles in the environment. All in all, their satisfaction is essential to an individual's

healthy development. Frustration of the secondary motives may result in emotional disturbances as well as physical illness because of lack of homeostasis. c) **Social motives**

- ❖ These are motives acquired through *social or cultural experience and interaction with others*. These include *affiliative* and *prestige* motives. These motives are found in social groups like: the peer groups, the family, religious groups and political groups. Within

which these groups, an individual strives to be liked, loved, admired, respected and envied.

- ❖ Socially oriented motives are classified in to two broad groups: *Affiliative* and *prestige* motives.

a) **Affiliative Motives**

- ❖ Affiliative motives represent the desire for love and belonging. Some examples of these motives are as follows:

- ✚ Desire to be with other people (gregariousness).
- ✚ Preference for social interaction with people having common interests
- ✚ Desire to have intimate contact with one's family.

- ❖ Affiliative motives are developed from early childhood and are modified as time goes on.

They play the following roles:

- ✚ They are helpful in *character development*. For example, through interaction with other people, a child is able to know which behaviour is socially acceptable and which is not.
- ✚ Affiliative motives also influence *moral learning* at home and at school. A child who behaves well is commended, one who errors is reprimanded. Therefore, children will do their best so as to get commendation from parents, teachers and peer group members.
- ✚ Affiliative motives *enhance language development*. Through interaction with other people, a child develops language, an essential mode of communication.
- ✚ Affiliative motives *enhance an individual's level of aspiration*. For example, children at school work better with the teacher that they like. Such a teacher should know when to commend or reprimand and when to use incentives. All these mould a child's attitude towards school, and therefore his aspirations.

b) Prestige Motives

- ❖ Prestige motives are those motives aimed at *enhancing the degree to which an individual is admired, respected or envied*. They are also known as *achievement motives, superiority motives and esteem motives*.
- ❖ Prestige motives are tied to an *individual's cultural environment*. They encourage one to strive to achieve a certain goal and to be recognized for it.
- ❖ Prestige motives are *responsible for the competitive spirit* found in any society in areas like fashion, business and education.

Relevance of Prestige and Affiliative Motives in a Learning Environment

Prestige motives

- Prestige motives encourage the competitive spirit found in any educational setup. Teachers should therefore *place their students in a competitive situation* as often as possible with a view to improving the quality of written and oral work.
- *Competitiveness should be varied* so that all students meet with success occasionally.

Affiliative motives

- Students should be *encouraged to identify with one another* and be loyal to the class as a group. The teacher should be perceived as a member of the group.
- The teacher should foster the affiliative spirit by *delegating authority and responsibility* in class and including class members in the planning of classroom activities. **Maslow**

Hierarchy Of Needs

- ❖ Maslow (1970) suggests that humans have a *hierarchy of needs* ranging from lower level needs to higher level needs.
- ❖ The *lower level* needs are *survival and safety related*.
- ❖ Maslow believed that humans have specific needs that must be met and that if lower level needs go unmet, we can not possibly strive for higher level needs.
- ❖ The Hierarchy of needs shows that at the lower level, we must focus on basic issues such as food, sleep, and safety.

- ❖ The *higher level* needs are those for *intellectual achievement and self-actualization*.

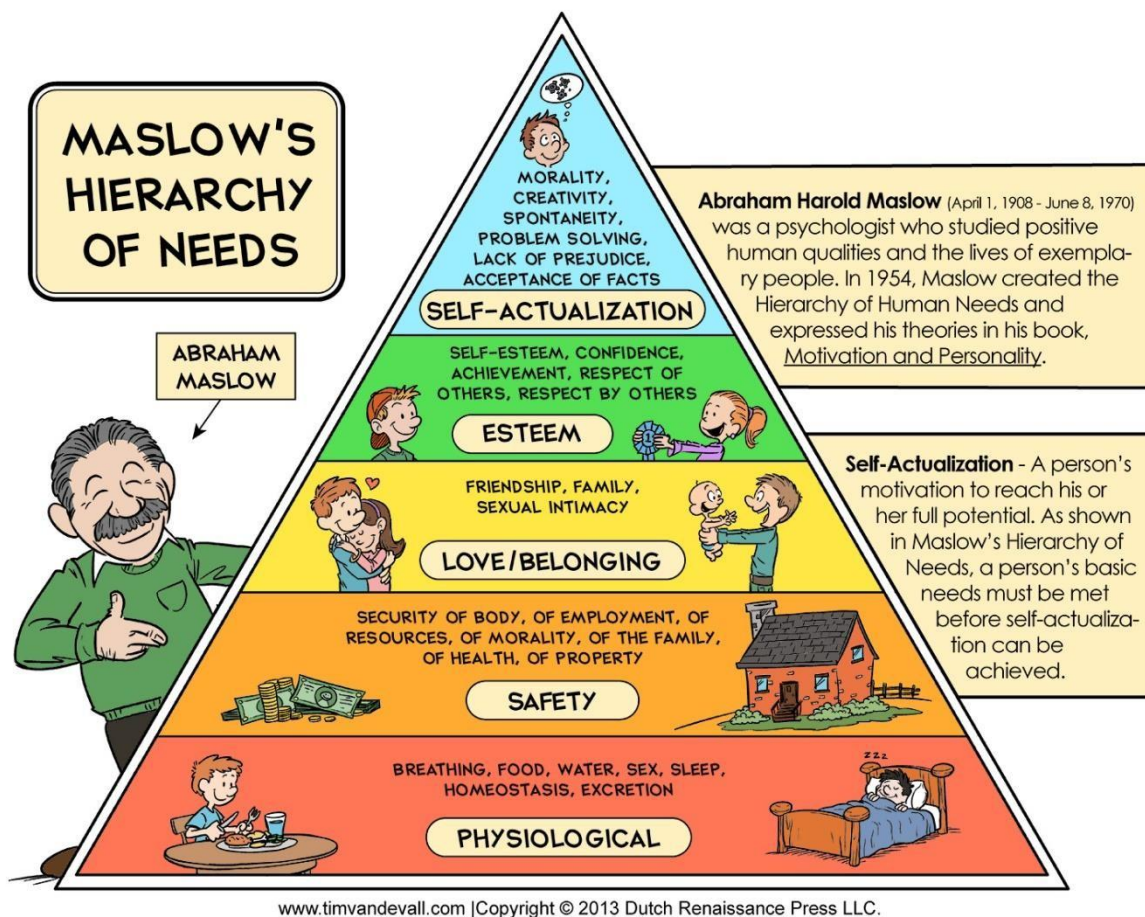


According to Maslow (1970), the five levels of the hierarchy have the following features:

- ✚ Level 1: ***physiological needs***, which are biological in nature and are unlearned. These needs must be satisfied before an individual can satisfy the needs pertaining to the next level. The physiological needs are: ***Need for food, water, oxygen and relief from pain***, among others.
- ✚ Level 2: encompasses ***safety needs***, which are person's desire to feel secure in nonthreatening physical and social environments which are learned through an individual's interaction with agents in the environment. Some of the safety needs include the need to ***save money, insure property, acquire job security and live in an atmosphere of peace***.
- ✚ Level 3: addresses ***love and belonging needs***. It is the need to be loved and accepted by significant others like. For example, the desire for ***gregariousness*** and ***identification with others***, in addition to the need to ***give and receive friendship and affection***.
- ✚ Level 4: covers the ***esteem needs***. This is the need to be outstanding in various aspects of life. They are aimed at ***enhancing a person's position*** in a group or society. Examples of esteem needs include the ***need to achieve success, competence and recognition in areas like business, politics, and academics***.
- ✚ Level 5: entails ***self-actualization*** or ***self-fulfillment***. This is the need to become the best one could be. It aims at developing the ***full potentialities of an individual***, in terms of creativity, self-expression, creativity, spiritual cleanliness, and community service. Some of the people who may have reached this level which transcends ordinary experience include:
 - Mother Theresa (Community Service).
 - Mozart (Music)
 - Jesus (Religion)
 - Mahatma Gandhi (Politics)
 - Martin Luther King (Civil Rights)

▪ Dos Santos Pele (Sports)

Throughout our lives, we work toward achieving the top of the pyramid, self actualization, or the realization of all of our potential. As we move up the pyramid, however, things get in the way which slow us down and often knock us backward. Imagine working toward the respect and recognition of your colleagues and suddenly finding yourself out of work and homeless. Suddenly, you are forced backward and can no longer focus your attention on your work due to the need for finding food and shelter for you and your family.



EMOTIONS



Introduction

According to Morris and Maisto (2005) the ancient Greek rationalists thought emotions, if not held in check, would wreck havoc on higher mental abilities such as rational thought and decision making.

- ❖ In the past, psychologists too viewed emotion as a base instinct-a vestige of our evolutionary heritage that needed to be repressed.
- ❖ More recently however, scientists have begun to see emotions in a more positive light.
- ❖ Today they are thought of as essential to survival and a major source of personal enrichment.

Definition of the concept

- ❖ A complex state of feeling that results in physical and psychological changes that influence thought and behavior.
- ❖ A complex feeling/state involving conscious experience and internal and overt physical responses that tend to facilitate or inhibit motivated behaviour.
- ❖ Outward expression of inner feelings.
- ❖ A motivated state marked by physiological arousal, expressive behaviour, and mental experience.
- ❖ For example –an angry man:
 - ✚ His heart might pound(physiological arousal)
 - ✚ Feel enraged(mental/cognitive experience) ✚
 - Grit his teeth(expressive behaviour)
- ❖ Kosslyn and Rosenberg (2006) propose that emotions have four components:
 - ✚ a positive or negative subjective experience ✚
 - bodily arousal

✚ the activation of specific mental processes and stored information, and ✚ characteristic behaviour



A feeling, or affect, that can involve physiological arousal (a fast heartbeat for example), conscious experience (thinking about being in love with someone, for example), and behavioural expression (a smile or grimace, for example).

- ❖ Emotion is often associated and considered reciprocally influential with mood, temperament, personality, disposition, and motivation. It also is influenced by hormones and neurotransmitters such as dopamine, noradrenaline, serotonin, oxytocin and cortisol.

Functions of emotions

- ❖ Help guide us to approach some things and withdraw from others
- ❖ Provide visible cues that help other people know key aspects of our thoughts and desires

Components of emotions

- **Cognitive appraisal:** provides an evaluation of events and objects
- **Bodily symptoms:** the physiological component of emotional experience
- **Action tendencies:** a motivational component for the preparation and direction of motor responses.
- **Expression:** facial and vocal expression almost always accompanies an emotional state to communicate reaction and intention of actions
- **Feelings:** the subjective experience of emotional state once it has occurred

Theories of emotion

Ancient Greece and Middle Ages

- ❖ Theories about emotions stretch back to at least as far as the stoics of *Ancient Greece and Ancient China*.
- ❖ In the latter it was believed that excess emotion caused damage to *qi*, which in turn, damages the vital organs.



The four humours theory made popular by Hippocrates contributed to the study of emotion in the same way that it did for medicine

- ❖ Western philosophy regarded emotion in varying ways. In stoic theories it was seen as a hindrance to reason and therefore a hindrance to virtue.
- ❖ Aristotle believed that emotions were an essential component to virtue.
- ❖ In the Aristotelian view all emotions (called passions) corresponded to an appetite or capacity.
- ❖ During the *Middle Ages*, the Aristotelian view was adopted and further developed by scholasticism and Thomas Aquinas in particular.
- ❖ There are also theories in the works of philosophers such as René Descartes, Niccolò Machiavelli, Baruch Spinoza and David Hume.
- ❖ In the 19th century emotions were considered adaptive and were studied more frequently from an empiricist psychiatric perspective.

Evolutionary theories



Illustration from Charles Darwin's *The Expression of the Emotions in Man and Animals*.

19th Century

- ❖ Perspectives on emotions from evolutionary theory were initiated in the late 19th century with Charles Darwin's book *The Expression of the Emotions in Man and Animals*.



- ❖ Darwin argued that emotions actually served a purpose for humans, in communication and also in aiding their survival.
Darwin, therefore, argued that emotions evolved via natural selection and therefore have universal cross-cultural counterparts.
- ❖ Darwin also detailed the virtues of experiencing emotions and the parallel experiences that occur in animals .
- ❖ This led the way for animal research on emotions and the eventual determination of the neural underpinnings of emotion.

Contemporary theories

The major theories of motivation can be grouped into three main categories:

- ✚ ***Physiological theories***- suggest that responses within the body are responsible for emotions.
- ✚ ***Neurological theories***- propose that activity within the brain leads to emotional responses
- ✚ ***Cognitive theories***- these theories argue that thoughts and other mental activity play an essential role in the formation of emotions.

Physiological theories

a) The James-Lange Theory of Emotion

- ❖ It was proposed by psychologist William James and physiologist Carl Lange.
- ❖ It suggests that emotions occur as a result of physiological reactions to events.
- ❖ Emotions occur when individuals become aware of a physiological response to an emotion-provoking stimulus.
- ❖ James believed that you feel emotions after your body reacts.
- ❖ For example, if a soldier raised his gun the moment he saw you coming, James would say that you would first run, and then feel afraid.



- ❖ In his 1884 article William James argued that feelings and emotions were *secondary* to physiological phenomena. In his theory, James proposed that the perception of what he called an "exciting fact" led directly to a physiological response, known as "emotion." To account for different types of emotional experiences, James

proposed that stimuli trigger activity in the autonomic nervous system, which in turn produces an emotional experience in the brain. The Danish psychologist Carl Lange also proposed a similar theory at around the same time, and therefore this theory became known as the James–Lange theory. As James wrote, "*the perception of bodily changes, as they occur, is the emotion*." James further claims that "we feel sad because we cry, angry because we strike, afraid because we tremble, and neither we cry, strike, nor tremble because we are sorry, angry, or fearful, as the case may be."

- ❖ An example of this theory in action would be as follows: An emotion-evoking stimulus (snake) triggers a pattern of physiological response (increased heart rate, faster breathing, etc.), which is interpreted as a particular emotion (fear).
- ❖ The issue with the James–Lange theory is that of causation (bodily states causing emotions and being *a priori*)
- ❖ The James–Lange theory has remained influential. Its main contribution is the emphasis it places on the embodiment of emotions, especially the argument that changes in the bodily concomitants of emotions can alter their experienced intensity.

b)The Cannon-Bard Theory of Emotion

- ❖ This theory suggests that the thalamus simultaneously sends an emotion-provoking stimulus to the brain providing feeling of an emotion, and to the sympathetic nervous system, causing physiological reactions such as sweating, trembling and muscle tension simultaneously.
- ❖ For example you perceive the threat of being shot, and the results of that perception marshal the body's resources for fleeing or fighting at the same time as they generate an emotion.
- ❖ Bodily emotions and the experience of emotion arise in tandem/simultaneously.

Cognitive theories

a)Schachter-Singer Theory/ the two-factor theory of emotion.

- ❖ This theory suggests that the physiological arousal occurs first, and then the individual must identify the reason behind this arousal in order to experience and label it as an emotion.
- ❖ Schachter and Jerome Singer (1962) developed a theory of emotion that gave cognition a greater role in emotion.
- ❖ They proposed that emotion is determined by two main factors:
 - ✚ Physiological arousal
 - ✚ Cognitive labeling
- ❖ They argue that we look to the external world for an explanation of why we are aroused.
- ❖ We interpret the external cues present and label the emotion.
- ❖ For example if you feel good after someone has made a pleasant comment to you, you might label the emotion —happy. If you feel bad after you have done something wrong, you may label the feeling —guilty.
- ❖ Schachter did agree that physiological reactions played a big role in emotions. He suggested that physiological reactions contributed to emotional experience by facilitating a focused *cognitive appraisal of a given physiologically arousing event* and that this appraisal was what defined the subjective emotional experience. Emotions were thus a result of two stage process: general physiological arousal, and experience of emotion.
- ❖ For example, the physiological arousal, heart pounding, in a response to an evoking stimulus, the sight of a lion in the kitchen. The brain then quickly scans the area, to explain the pounding, and notices the lion. Consequently, the brain interprets the pounding heart as being the result of fearing the lion.

b)Richard Lazarus

- ❖ Lazarus (1991) believes cognitive activity is a precondition for emotion. ❖ He says we cognitively appraise ourselves and our social circumstances.
- ❖ These appraisals which include values, goals, commitments, beliefs, and expectations determine our emotions.
- ❖ People may feel happy because they have a deep religious commitment, angry because they did not get the raise they expected, or fearful because they expect to fail an exam.

- ❖ Richard Lazarus argued that *emotions must have some cognitive intentionality*. The cognitive activity involved in the interpretation of an emotional context may be conscious or unconscious and may or may not take the form of conceptual processing.
- ❖ Lazarus' theory is very influential; emotion is a disturbance that occurs in the following order:
 - ✚ **Cognitive appraisal**—the individual assesses the event cognitively, which cues the emotion.
 - ✚ **Physiological changes**—the cognitive reaction starts biological changes such as increased heart rate or pituitary adrenal response.
 - ✚ **Action**—the individual feels the emotion and chooses how to react.

For example: Jenny sees a snake.

1. Jenny cognitively assesses the snake in her presence. Cognition allows her to understand it as a danger.
2. Her brain activates Adrenaline gland which pumps Adrenaline through her blood stream resulting in increased heartbeat.
3. Jenny screams and runs away.

Behavioral dimension

- ❖ The facial feedback hypothesis states that facial expressions can influence emotions as well as reflect them.
- ❖ In this view, facial muscles send signals to the brain, which helps individuals recognize the emotion they are experiencing.
- ❖ Particular facial expressions alter the flow of blood to particular regions of the brain, thereby evoking particular emotional experiences.

Sociocultural dimension

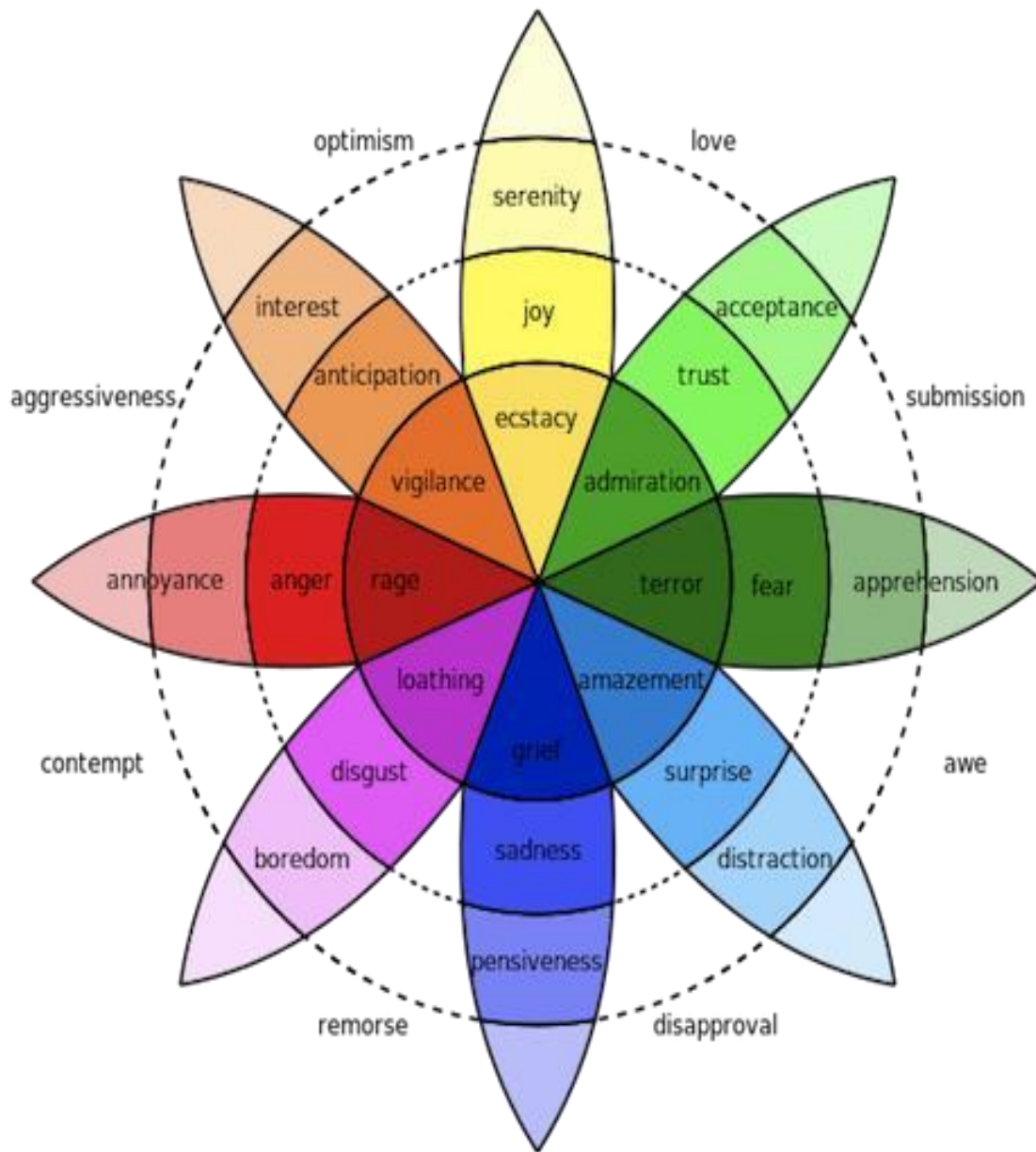
- ❖ Paul Ekman's (1980,1996) careful observations reveal that our many faces of emotion do not vary significantly from culture to culture.

Plutchik's Eight Basic Emotions

Plutchik's eight **basic** or **primary** emotions are **Joy**, **Trust**, **Fear**, **Surprise**, **Sadness**, **Anticipation**, **Anger**, and **Disgust**. Each primary emotion has a polar opposite, so that:

- *Joy* is the opposite of *Sadness*
- *Fear* is the opposite of *Anger*
- *Anticipation* is the opposite of *Surprise*
- *Disgust* is the opposite of *Trust*

Plutchik created the **wheel of emotions** in order to illustrate the various relationships among the emotions.



NB:

- ✚ The intensity of emotion decreases as you move outward and increases as you move toward the wheel's center.
- ✚ The intensity of the emotion is indicated by the color.
- ✚ The darker the shade, the more intense the emotion.

- ✚ For example, anger at its least level of intensity is annoyance. At its highest level of intensity, anger becomes rage.

FRUSTRATION AND CONFLICT

Definition of Frustration

- ❖ Frustration is the *blockage of motives by obstacles*. Frustration may also be the delay or interference with the attainment of a goal (heavy traffic jam).
- ❖ Experienced whenever the *pursuit of a goal is thwarted/impended* or when we are *not able to satisfy a motive*.
- ❖ *Negative emotional state* that occurs when *people are prevented from reaching desired goals*.
- ❖ Frustration may be threatening or non-threatening. For example, a faulty light bulb may lead to the mild inconvenience of looking for a spare one, while losing money to thieves may be threatening, if one wanted to use the money to pay a hospital bill.

Sources of Frustration

- Environmental obstacles
- Personal obstacles
- Conflicts

Environmental Obstacles

- ❖ Our physical environment may cause frustration through serious obstacles like earthquakes, tornadoes, famine and floods, among others. Some petty obstacles provided by the environment include traffic jams, tyre burst, and disturbance by barking dogs at night, loud music at night, uneven sidewalks and rain at picnic sites.
- ❖ The social environment provides obstacles like: rules and regulations that are perceived as being too strict; poor working conditions, racial prejudice and tribalism.

Personal Obstacles

An individual may suffer from frustration if some personal limitation/inadequacy prevents the satisfaction of various motives. These limitations may be real or imagined. Some of the limitations are:

- lack of skills in games
- lack of intellectual potential
- lack of economic wealth
- excessive fatness or shortness
- short sight/long sight, blindness
- unusual skin pigmentation (Albinism)
- Paralysis or other physical disability yet wants to dance
- Unattractiveness
- Shyness
- Chronic illness

Conflicts

- Occurs when *two or more incompatible motivations* or behavioural impulses *compete for expression*.
- When we face two or more incompatible demands, opportunities, needs or goals. We end up:
 - giving up some of our goals
 - modifying some of them
 - resigning ourselves to not attaining all of our goals
- Our inability to decide on the next course of action in view of the fact that every day, we all come across many situations in which we must make decisions, or select an alternative
- If the state of conflict persists, we often develop frustration e.g. having to decide whether to go to the University for a Degree or getting married; having to choose between two good girlfriends/boyfriends to marry.

Types of conflicts

Psychologists have classified conflicts into the following four categories:

a) *Approach – Approach Conflict*

- ❖ This type of conflict is also known as ***conflicting attraction***.
- ❖ It results from the existence of two or more equally attractive goals.
- ❖ These goals are ***mutually exclusive***, that is, we must choose one at the expense of the other.
- ❖ In this kind of conflict, the period of indecision is dependent on the degree of goal equivalence, in that the longer the period of indecision, the greater the amount of stress and frustration.
- ❖ Most individuals will move from one goal to the other in an attempt to evaluate them. This process is called ***vacillation***. It is the perceived importance of either goal that helps to decide.

In order to escape this conflict, an individual must do the following:

- A decision must be made promptly regarding one of the alternatives, making sure that no feelings of frustration are left behind.
- In some cases, it is possible to satisfy one goal at once, leaving the other goal open for satisfaction at a later date.

b) Approach – Avoidance Conflict

- ❖ A choice must be made about whether to ***pursue a single goal*** that has ***both attractive and unattractive aspects***.
 - For example, a journalist may enjoy his work, but may fear the prospect of being injured while covering a story during university students' riot; risking rejection by approaching an attractive person in class; risking your savings by investing in a new business that could fail; wanting to join a high status group but can do so only by endorsing views contrary to one's values.
- ❖ An individual needs to make a decision promptly to minimize stress. The decision will involve acceptance of the consequences of one's decision

c) Avoidance – Avoidance Conflict

- ❖ This conflict occurs when an individual is faced with two unpleasant alternatives.
- ❖ In this case, it is impossible to avoid one alternative without experiencing the other.
- ❖ One confronts ***two undesirable or threatening possibilities***, neither of which has any positive attributes i.e. —caught between a rock and a hard place
- ❖ e.g. a person with a toothache must choose between the pain of the tooth and the anticipated discomfort of going to the dentist; choosing between finishing a job one intensely dislikes or quitting and being called a failure; a person who has been admitted to the university to pursue a course he does not like- the alternative is to pursue the course of his choice in a less prestigious institution; fear of punishment in school vs. fear of punishment at home etc.
- ❖ Values and one's character are required to resolve the conflict.
- ❖ Some people resort to **defense mechanisms** to postpone making a decision, simply because all the alternatives are unpleasant.

Note:

Defense mechanisms are the reactions people develop when they wish to avoid making decisions. For example, some people may fantasize that the issue does not exist.

d) Double Approach – Avoidance Conflict

- ❖ This type of conflict involves two or more alternatives, each of which has both attractive and repulsive qualities- ***each alternative has both negative and positive qualities***
- ❖ For example, you are offered two jobs; one has good pay but poor hours and dull work; the second has interesting work and excellent hours but low pay; a person may get an offer of a highly paying job abroad but is not allowed to go with his family- the alternative is to stay with his family and continue with the unattractive local job.
- ❖ An individual must make his choice after careful analysis of the consequences of his decision.
- ❖ This conflict is difficult to cope with because one is never sure that the correct choice has been made.

How do people react to frustration?

(Differently due to type of personality, alternatives, level of stress etc)

- ❖ Withdrawal
- ❖ Hostile feelings directed at the wrong people/scapegoating
- ❖ Poor grooming
- ❖ Alcoholism
- ❖ Workaholics
- ❖ Excessive smoking
- ❖ Vulgar expressions
- ❖ Over religiosity/subservience
- ❖ May also lead to creativity (new ways of doing things) for those with a strong will.
- ❖ Use of defense mechanisms

Defense mechanisms

We are sometimes forced to use delaying tactics in order to avoid making decisions. These tactics are the defense mechanisms. There are short-term and long-term defense mechanisms.

Short-term Defense Mechanisms

1) Restlessness and Tension

- ❖ The emotional reactions associated with restlessness and tension includes whimpering, sighing and complaining. These are immediate symptoms of frustration.
- ❖ Should the amount of stress within the individual escalate, other symptoms exhibited include trembling, clenching of fists, smoking, chewing gum, nail biting and thumb sucking.

2) Destructiveness

They are of two types:

- Direct aggression: Frustration may lead to destructiveness as well as hostile attacks on all persons and things that are seen causing frustration. Direct aggression involves kicking out, knocking over or breaking and destroying objects.
- Indirect aggression: There are two forms of indirect aggression: ***displaced aggression and scape-goating***. Displaced aggression involves aggression against innocent persons or objects rather than the actual cause of frustration. For example, a frustrated employee may harass his family instead of his domineering boss. **Scape-goating** involves placing blame on an innocent person for one's shortcomings. For example, a student blames his teacher for his low grade.

3) **Apathy**: Apathy is a state manifested through the following:

- + Indifferences
- + Inactivity
- + Inattentiveness
- + A general don't care attitude

NB: Prisoners resort to apathy when they realize that aggression against detention is futile behaviour.

4) **Fantasy**:

- ❖ This is the tendency to resort to unrealistic solutions.
- ❖ Fantasy usually involves escapism into the dream world by both adults and children.
- ❖ In extreme cases, individuals may lose ability to distinguish between reality and the fantasy world.

5) **Stereotype**:

- ❖ This is the tendency to resort to repetitive, fixed behaviour.

- ❖ This behaviour is repeated despite its failure to solve problems in previous efforts, for example, thumbing in children.

6) Regression:

- ❖ It involves a return to primitive modes of behaviour that were present at a younger age. It is often characterized by an attempt to return to a more desirable past stage of development. Disorganization and collapse of adult behavior and emergence of developmentally lower levels of functioning.
- ❖ A retreat to an earlier developmental level according to Freud – the person's most fixated stage that demands less mature responses and aspirations.
- ❖ Return to the use of reaction patterns long since outgrown in the face of adult difficulties yearning for the carefree, sheltered days of childhood e.g The frustrated adult may resort to temper tantrums of childhood; the bride may run home to mother her at the first sign of trouble; tendency of old people to live more and more in the past and extol its advantages —the good old days; in the face of severe stress, an individual may retreat from adult reaction patterns to a less demanding status that involves lowered aspirations and more readily accomplished satisfactions; psychotic adults who show such extreme regression to infantile levels of behavior that they are unable to wash, dress, or feed themselves, or take care of their eliminate needs. In some cases, they curl up in a position similar to that of the foetus in the womb

Long Term Defense Mechanisms.

1) Rationalization:

- ❖ This involves giving logical reasons or excuses for our behaviour rather than the real/true reason for an action or event / Involves thinking up logical, socially approved reasons for past, present or proposed behavior
- ❖ For example, a person tells his friend that he didn't get a job because he had no connections; justifying oneself for spending money needed for essentials or lavish entertainment & neglecting work for other pursuits; marriage proposal is turned down by an attractive girl (after all she talks too much, and will probably lose her figure at an early age); a person has very little money- emphasizes that the really important things in

life- such as love and friendship are free; mediocre performances e.g. test failure- —I am not interested in the course. Besides, I find the teacher extremely dull. I do not care about grades anyway.‖

2)Projection:

- ❖ This involves attributing one's own undesirable traits or impulses to other people. Places the blame for his own shortcomings, mistakes and mistakes on others.
- ❖ Attributes to others his own unacceptable impulses, thoughts and desires (ascribing feelings to someone else that we don't want to acknowledge as our own.)
 - E.g. A student, who feels angry with the teacher, perceives the teacher as being angry with him; the delinquent teenager may blame his problems on a rejecting and non – understanding mother; a lonely divorced woman accuses all men of having only one thing in their minds.

3)Reaction formation:

- ❖ It involves expressing exaggerated ideas and emotions that are the opposite of the disturbing motive, unconscious impulses or desires/ Repression of dangerous impulses, followed by converting them to their direct opposite
- ❖ For example, a father may show excessive love to a child he does not like in order to cover up his real feelings; a former prostitute Angel is now a tireless crusader against it; a woman is attracted to a man, but acts coldly towards him concealing hate with a façade of love; a woman who gives birth to an unwanted child becomes extremely overprotective; people who crusade against loose morals, alcohol, gambling, and other real or alleged evils- often such people have a background of earlier difficulties with these problems themselves and their zealous crusade appears to be a means of safeguarding themselves against recurrence of their difficulties.

4) Dissociation:

- ❖ It is a mechanism that disrupts the unity of thinking, feeling and acting.
- ❖ One form of dissociation is compulsive behaviour that is characterized by actions that are not well-coordinated.
- ❖ A twitching arm may substitute for hitting out in anger.

❖ Dissociation keeps away feelings of guilt by automatically suppressing undesirable acts.

5)Substitution: It involves substitution of approved goals for unapproved ones. There are two forms of substitution, *sublimation* and *compensation*.

✚ Sublimation: This is a process where socially unacceptable motives are rechanneled into pursuits that the society considers acceptable or admirable. For example, the desire to express aggressions can be channeled to sports or occupation like boxing, and rugby. Sexual energy can be channeled to art and music.

✚ Compensation:

- ✓ This is a process an individual strives to make up for failure in one activity by making an effort in another activity.
- ✓ A defense against feelings of inferiority and inadequacy growing out of real or imagined personal defects or weaknesses, as well as out of the individual's inevitable failures and setbacks
- ✓ For example, a student who is poor in sports may focus his energy into academic work; Theodore Roosevelt waged a valiant fight against ill health to become noted for his physical daring and robustness; physically unattractive person may develop an exceptionally pleasing personality; the boy who feels inferior and unpopular may become the local bully; some people resort to criticism or innuendos in an attempt to cut others down to their own size; engaging in antisocial behavior in an attempt to get attention.

6)Repression:

- ❖ This is a process where stressful things are conveniently forgotten.
- ❖ Unpleasant memories or disturbing sexual and aggressive impulses are involuntarily removed from consciousness or barred from consciousness.
- ❖ For example, after failing an exam, you keep forgetting to tell your parents about it; forgetting to go to a dreaded dentist appointment; a soldier who witnesses the horrible death of his friend in combat becomes 'amnesic' with regard to the battle experience; Job forgets a traumatic incident from childhood.

7)Intellectualization:

- ❖ This is an attempt to gain detachment from an emotionally threatening situation.
- ❖ Threatening thoughts or emotions are kept at arm's length by thinking about them logically and rationally
- ❖ For example, the hurt over a parent's death is reduced by saying that he/ she lived a full life or died mercifully without pain; failures and disappointments are softened by pointing out that, —It could have been worse.‖

8)Suppression:

- ❖ The process of deliberate self-control. It involves keeping impulses or desires in check.
- ❖ It also involves consciously keeping aside painful memories in order to concentrate on other tasks. **9)Denial-**

- ❖ refusing to accept some aspect of reality because it creates too much anxiety
- ❖ An attempt to —screen – out‖ disagreeable realities by ignoring or refusing to acknowledge them' —This isn't really happening to me.‖e.g a smoker refuses to acknowledge the risks of smoking; refusing to discuss unpleasant topics ,fainting when confronted with a traumatic situation, ignoring criticism, becoming pre – occupied with work, hence having no time e.g. to deal with personal problems; one has a drinking problem, but denies that it is a problem.



10)Identification-

- ❖ Incorporating someone else into our own personality, adopting their behaviour and attitudes.
- ❖ Taking on the characteristics of someone else so that we can vicariously share in that person's triumphs and overcome feelings of inadequacy.
- ❖ People who feel inferior may identify themselves with successful causes and people or groups in the hope that they will be perceived as worthwhile.
- ❖ Enhances feelings of worth and protects the individual against self – devaluation e.g.a parent with unfulfilled career ambitions may share emotionally in a son's / daughter's professional success; adolescence – group identification; adults – identify themselves with their professions, homes, special interest groups and the accomplishments of their children; many

employees identify themselves with the power and prestige of the company for which they work.

11)Introjections

- ❖ Involves the acceptance of other's values and norms as one's own even when they are contrary to one's previous assumptions e.g. Introjection of Nazi norms in the German concentration camps; an abused child who assumes the abusing parent's way of handling stress and thus continuing the cycle of child beating.
- ❖ Note: Introjection can lead to seriously distorted and maladaptive behavior.

12)Undoing (Atonement)

- ❖ A symbolic attempt, often ritualistic or repetitive, to right a wrong or negate some disapproved thought, impulse or act e.g.
 - Apologizing for wrongs committed against others.
 - Penance, repentance and undergoing punishment (resolve guilt feelings and make a new beginning).
 - The unfaithful husband may bring his wife presents.
 - The unethical businessman may give huge sums of money to charity.
 - Dishonest businessman – who may be a kind father and a —pillar of the church).

13)Displacement

- ❖ Directing an emotion such as hostility or anxiety towards a substitute target e.g.
 - The meek office clerk who is refused a raise by his domineering boss. Instead of expressing his hostility towards his employer, he goes home and snaps irritably at his wife because supper is a few minutes late.
 - Sometimes one may turn the hostility inwards (intropunitive) engage in exaggerated self – accusations and recriminations and feel severely guilty (can lead to depression and suicide).
 - In Nazi Germany, the blame for the country's ills was attributed to the Jews and Communists/ scapegoating.
 - Swearing – used as a means of discharging pent – up feelings.
 - Destructive criticisms and vindictive gossip - disguised methods of expressing hostility.

NB: The above defence mechanisms help one to cope with frustration. If there is too much frustration, stress sets in and individuals may become maladjusted or get mental illnesses such as psychosis (madness) and neurosis

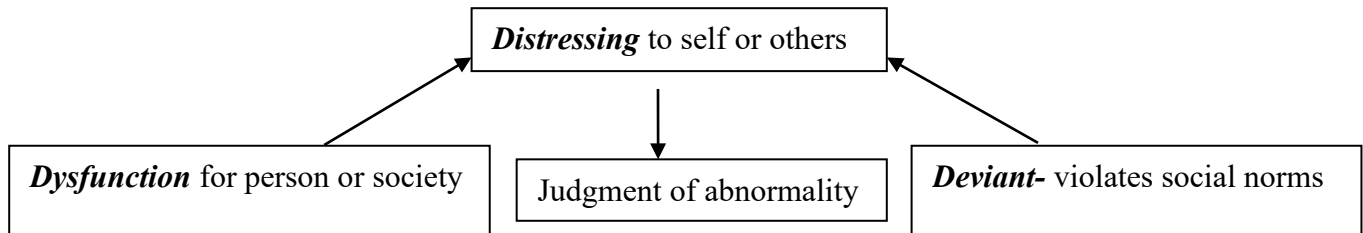
PSYCHOLOGICAL DISORDERS

- ❖ *Mental processes and/or behaviour patterns that cause emotional distress and/or substantial impairment in functioning.*
- ❖ Psychological disorders are associated with abnormal behaviours.
- ❖ The ancient *Chinese, Egyptians, and Hebrews* attributed abnormal behaviour to the work of the devil. To release the spirit, they carried out a procedure called *trephination* in which a hole was chiseled through the skull to release the evil spirit.
- ❖ In medieval Europe, the *demonological model* of abnormality reigned supreme. The killing of witches was justified on theological grounds. Witches were thrown into a lake or pond.
- ❖ In the *5th century BC*, the *Greek physician Hippocrates* suggested that mental illnesses are a disease like any other. He believed that the seat of mental illness was the brain. His belief that a mental disorder could be caused by a physical dysfunction is today reflected in the biological perspective on psychological disorders.
- ❖ By the *1800's*, *Western medicine* had returned to viewing *mental disorders* as being *biologically based*, and was attempting to extend *medical diagnosis* to them.
- ❖ From the *early 1900's*, other models of abnormal behaviour were ushered in e.g. *psychoanalysis, behavioral, cognitive and humanistic conceptions*.

Defining Abnormality

- **Lahey (2004)** defines abnormal behavior as *actions, thoughts and feelings that are harmful to the person or to others*
- **Kosslyn (2004)** opines that a psychological disorder is signaled by a *constellation of cognitive, emotional, and behavioral symptoms that create significant distress; impair work, school, family, relationships, or daily living, or lead to significant risk of harm*.
- **Diagnostic and Statistical Manual of Mental Disorders (DSM-IV)** defines abnormality as:

“a clinically significant behavioral or psychological syndrome or pattern that occurs in an individual and that is associated with present distress (e.g. a painful symptom) or disability (i.e. impairment in one or more important areas of functioning) or with a significantly increased risk of suffering death, pain, disability, or an important loss of freedom”



*NB: Whether a behavior is considered **abnormal** involves a social judgment made on the basis of the “three Ds,”: distress, dysfunction and deviance.*

Explaining abnormality/The etiology of abnormality/Models of abnormality

i. The biological/Medical model

- ❖ It holds that abnormal behavior is caused by *physiological malfunctions* e.g. of the nervous system or the endocrine glands-often stemming from hereditary factors/hormonal imbalances/chemical deficiency/brain injury.
- ❖ There is growing evidence that genetic factors are involved in mental disorders as diverse as schizophrenia, depression, and anxiety and that the biochemistry of the nervous system is linked to some cases of depression and schizophrenia.
- ❖ The cause of abnormality can thus be found through a physical examination of the individual.

ii) The psychoanalytic model

- ❖ According to this model, behavior disorders are symbolic expressions of *unconscious conflicts*, which can usually be *traced to childhood traumas or anxieties* e.g. a man who behave towards women in a violent way may be unconsciously expressing rage at his mother for being unaffectionate toward him during his childhood

- ❖ The psychoanalytic model argues that in order to resolve their problems effectively, people must become aware that the source of their problems lies in their childhood and infancy.

iii) *The behavioral model*

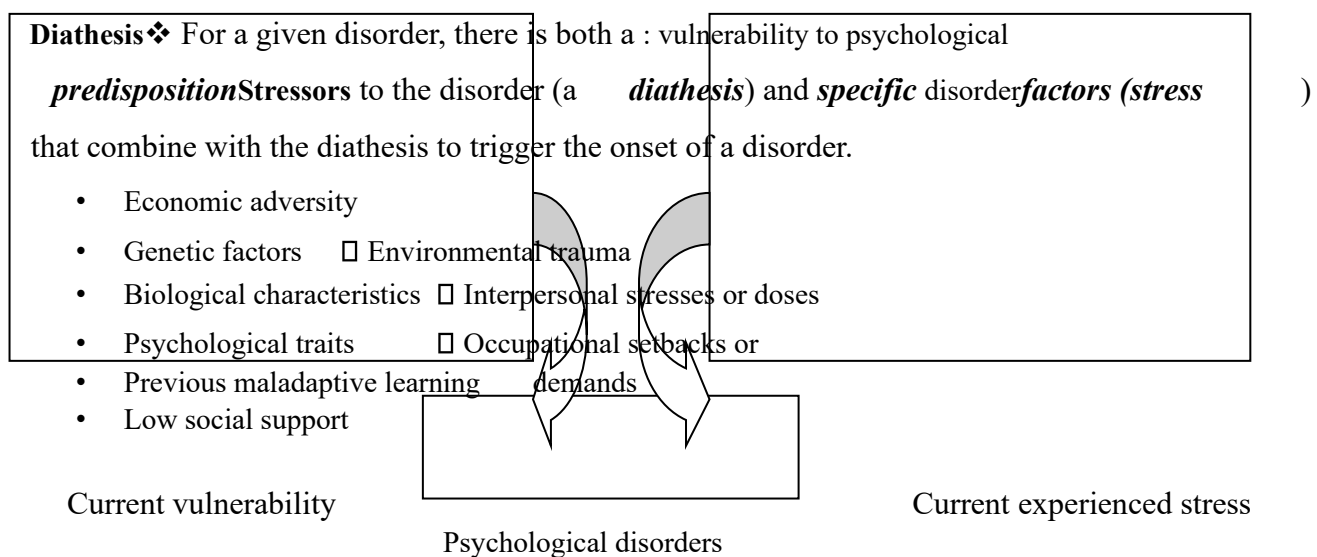
- ❖ It suggests that abnormal behavior results from **learning/product of learning**.
- ❖ It is in **response** to a set of **stimuli** in the individual's environment.
- ❖ From this perspective, fear, anxiety, sexual deviations and other maladaptive behaviors are learned and can be unlearned.

iv) *The cognitive model*

- ❖ Cognitions (people's thoughts and beliefs) are central to a person's abnormal behaviour
e.g. —Doing well in this exam is crucial to my entire future.‡
- ❖ Ineffective or inaccurate thinking is the root cause of mental illness /faulty cognitive style/maladaptive cognitive processes.

v) *The Diathesis – stress model*

- ❖ It suggests that a **biological predisposition** called a **diathesis** must combine with some kind of **stressful circumstance** before the predisposition to abnormality shows up as a behavior
- ❖ According to this model, some people are **biologically prone to developing a particular disorder under stress**, whereas others are not.



Vulnerability-stress model views behaviour disorders as resulting from an interaction between ***personal vulnerability*** factors and ***life stressors***. Personal vulnerability factors contribute to maladaptive efforts to cope with life's challenges.

vi) The sociocultural perspective

- ❖ It proposes that mental illness is the product of broad social and cultural factors;
 - ✓ eg disorders are higher in poor urban settings
 - ✓ due to unemployment
 - ✓ poverty
 - ✓ inequality etc
- Franzoi(2006) indicates that:
 - ✓ There is a culture-bound syndrome- the Mediterranean cultures suffer from *mal de ojo* ' or evil eye in which they experience fitful sleep, unexplained crying, diarrhea, vomiting and fever.
 - ✓ Arctic and sub-arctic Eskimos –suffer from *pibloktoq* ' disorder, which involves abrupt break with reality, violence, hyperexcitability, followed by seizures and a coma.

vii) The family systems model

- ❖ Emphasizes the ***family's influence*** on individual behavior
- ❖ This viewpoint holds that all members of a family are enmeshed in a network of interdependent roles, statuses, values and norms.
- ❖ The behavior of one member directly affects the entire family system.
- ❖ Correspondingly, ***people typically behave in ways that reflect family influences***
- ❖ Personality development is ruled largely by the ***attributes of the family***, especially the way parents behave toward and around their children.
- ❖ ***Abnormal behavior*** in the individual is also a reflection or a ***—symptom” of unhealthy family dynamics***, and more specifically, of poor communication among family members (faulty and distorted communication)

- ❖ The *locus of disorder* is seen to reside not within the individual, but *within the family*

Neurosis

- ❖ Neurosis is a mild form of a mental disorder that causes people *considerable personal distress and some impairment in functioning*, without causing them to lose contact with reality or to violate important social norms.
- ❖ This condition usually results from inadequate adjustment responses, leading to a personality that is either too exaggerated or disorganized.
- ❖ In general, neurotics are unhappy individuals whose reaction patterns are inadequate for dealing with many life situations.
- ❖ It requires medical intervention. Neurosis may be severe to the point of incapacitating a person or it may be mild and unnoticeable to the individual.
- ❖ 1 in 20 people experience neurosis of the mild type, which does not require hospitalization (Floyd, 1988).

Some examples of conditions that are classified under neurosis are as follows:

1) *Anxiety Neurosis*

- ❖ This is characterized by a *general feeling of worry and apprehension*. Some people develop what is known as *free-floating anxiety*, where they feel afraid but cannot pinpoint the real cause. The *worrying* is *unrealistic* – there is little or any chance that the feared event will occur. Anxiety neurosis is often accompanied by:

- ✚ Tenseness, sweating and nausea;
- ✚ Inability to relax
- ✚ Muscle tension
- ✚ Rapid heartbeat or pounding of heart
- ✚ Apprehensiveness about the future
- ✚ Constant alertness to potential threats
- ✚ Sleeping difficulties

2) *Evasion of Growth:*

- This is the process where the *neurotic tends to think of himself as inadequate and unworthy of love and praise.*
- The individual is therefore unable to fulfill his potentialities and will eventually repress his individuality by acting like some person he admires, for example a parent, national hero or a friend. This is known as *vicarious* (indirect) *living*, which in turn makes the person feel emptier. Such an individual becomes non-committal in life and will probably not be able to keep a job for long.

3) Phobias:

- This is the *irrational fear of certain objects, people or situations.*
- Phobias are usually developed when free floating fear settles on a particular object, situation or people, leading to panic attacks when these stimuli present themselves. Some examples of phobias are as follows:

Examples of phobias

Phonophobia- speaking out loud

Aerophobia- flying

Ailurophobia- cats

Anthophobia- flowers

Aquaphobia- water

Astraphobia- lightning

Claustrophobia- closed places

Dementophobia- insanity

Herpetophobia- reptiles

Microphobia -germs

Pathophobia- fear of disease

Nyctophobia- darkness

Ophidiophoba- snakes

Pyrophobia- fire

Trichophobia- hair

Algophobia- fear of pain

Acrophobia- heights

Agoraphobia- open/public places

Amazophobia- vehicles/driving

Anthrophobia- people

Arachnophobia- spiders

Brontophobia- thunder

Cynophobia- dogs

Gephyrophobia- bridges

Hydrophobia- water

Murophobia- mice

Numerophobia- numbers

Ochlophobia- crowds

Ornithophobia- birds

Thanatophobia- death

Xenophobia- strangers

Monophobia- fear of being alone

4) Hysteria:

- This is appearance of *physical symptoms that have no organic origin*. For example, hysteria may be *associated with uncontrolled laughing or crying*.
- Some people may develop *temporary paralysis as a consequence of this condition*. Hysteria is usually expressed when individuals are under *severe stress*, which may be brought about by *pressure to achieve new or very advanced goals*. For example, some teacher-trainees on teaching practice may be hysterical on noticing the presence of the supervisor.
- Hysteria can also be expressed in terms of loss of sensitivity to touch and pain, or even blindness. That is why hysteria is often referred to as a *conversion reaction* because it involves conversion of stress into the body disturbances mentioned above.

5) *Obsessive – Compulsive Reactions:*

- The term ‘*Obsession*’ means that a person develops *persistent, irrational thoughts* that are usually unpleasant and cannot be banished voluntarily. For example, such an individual will keep on wondering whether he locked the main door to his house at night, and these recurrent thoughts will interfere with sleep.
- Obsessive thoughts are harmful because the individual will not be able to concentrate on other more proactive thoughts.
Closely associated with the ‘*Obsession*’ is ‘*compulsion*’; this term means that a person develops a *ritualistic behaviour, or a pre-occupation with regard to irrational thoughts*. A compulsive person will display behaviours like waking up many times at night to check whether the already locked door is indeed locked.
- The two concepts: ‘*Obsession*’ and ‘*Compulsion*’ are usually combined to make the concept ‘*Obsessive-compulsive reactions*’. This is because obsessions and compulsions usually occur together.

6) *Neurasthenia:*

- This refers to the presence of *feelings of fatigue and unpleasant physical sensations* like ringing in the ears and heart palpitations.
 - Neurasthenia originates from long-lasting emotional frustration, which drains a person’s energy.
- 7) *Hypochondriacs:*

- This is the ***tendency to show excessive concern to one's health***.
- The hypochondriac is therefore extremely alarmed about every minor bodily sensation. For example, a headache can be easily interpreted as the symptom of a tumor.
- Hypochondriacs frequent health centers and are avid readers of medical literature.
- They also brag about their imaginary medical milestones.

8) Dissociative Reactions:

- This term refers to the ***sudden repression of entire episode of one's life***, from the consciousness level.
- Some of the common example of dissociative reactions are:
 - ✚ ***Amnesia***, which involves loss of memory. The person is ***unable to remember important personal information***. The amnesia can be general e.g. failure to remember one's identity or family history, or it can be localized such as the failure to remember a specific traumatic life experience.
 - ✚ ***Fugue***, which in Latin means 'to flee', and is exhibited through a person wandering off to another place unconsciously. Loss of ***personal identity- people forget who they are***-and its accompanied by an ***escape or flight from the home*** environment (fugue – literally means flight). Such individuals ***adopt new identities*** in their new locale. They typically claim they have no knowledge of their activities during their blackout period.
 - ✚ ***Multiple Personality***, in which a person develops two or more distinct personalities and each personality, is not aware of the existence of the other. A person alternates among what appears to be ***two or more distinct identities*** or personality states. The individual's original personality is often conventional, moralistic and unhappy whereas the alternative personalities tend to be quite the opposite.

Psychosis

- This is a ***mental illness that is severe that the individual loses touch with reality***.
- Psychotics must be placed under institutional care to facilitate the management of this state.
- Some forms of psychosis develop suddenly, while others develop gradually.

- Psychotics do not exhibit denial of reality or denial of the existence of conflicts as in neurotics.
- Many psychotics are not able to distinguish between fantasy and reality, and that is why they engage in weird behaviour.
- Psychosis results from either *organic* causes or *functional* causes.
- *Organic causes* include *problems in the Central Nervous System (CNS)*, while *functional* causes include the *effects of stress*.

Symptoms of Psychotic Behaviour

- **Hallucinations:** These are *sensory perceptions in the absence of any external sensory stimulus*. i.e. it is an imaginary sensation.
- **Delusions:** They are *strong false beliefs*, not generally shared by others in the culture that cannot be changed despite strong evidence on the contrary. For example;
 - ✚ *Delusions of Grandeur:* A false belief that one is a famous person with great success, power, or ability.
 - ✚ *Delusions of Reference:* Misunderstanding chance occurrences
 - ✚ *Delusions of Persecution:* Perceptions that enemies are out to do harm.
 - ✚ *Emotional distortion:* Exaggerated reactions with respect to emotions and behaviour.
- **Disturbances in form of thought or speech-** derailment- *one does not follow one line of thought to completion*, but on the basis of vague connections shifts from one subject to another. Their speech is often difficult or even impossible to understand.
- **Inappropriate effect-** a person's *behaviour* e.g. their facial expressions, tone of voice and gestures does *not reflect the emotion that would be expected under the circumstances*. E.g. crying when watching a comedy or laughing when watching news on a fatal accident.

Example of Psychosis

1) Manic-depressive reactions:

- ❖ *These are extreme changes of mood from strong excitement to extreme sadness.*
- ❖ Characteristics:
 - ✚ High mental/Physical energy

- + Elation, restlessness
- + Laughter, aggression
- + Low energy
- + Sadness
- + Suicidal tendencies 2)

Schizophrenic reactions:

- ❖ The term schizophrenia means „*out of touch*“.
- ❖ The individual experiencing schizophrenia is not in touch with reality in his mental functioning.

Types of schizophrenia

- **Paranoid schizophrenia**- a type of schizophrenia characterized by *delusions of grandeur, reference and persecution in addition to hallucinations, and exaggerations of emotions*. Many paranoid people look perfectly normal.
- **Disorganized schizophrenia**- a serious type of schizophrenia characterized by *inappropriate effect, silliness, laughter, grotesque mannerisms, and bizarre behaviour*.
- **Catatonic schizophrenia**- a type of schizophrenia characterized by *complete stillness or stupor and/or periods of great agitation or excitement*. Patients may assume unusual posture and remain in it for long periods.
- **Undifferentiated schizophrenia**- a category used when *schizophrenic symptoms do not either conform to the criteria of any schizophrenia type* or conform to more than one type.